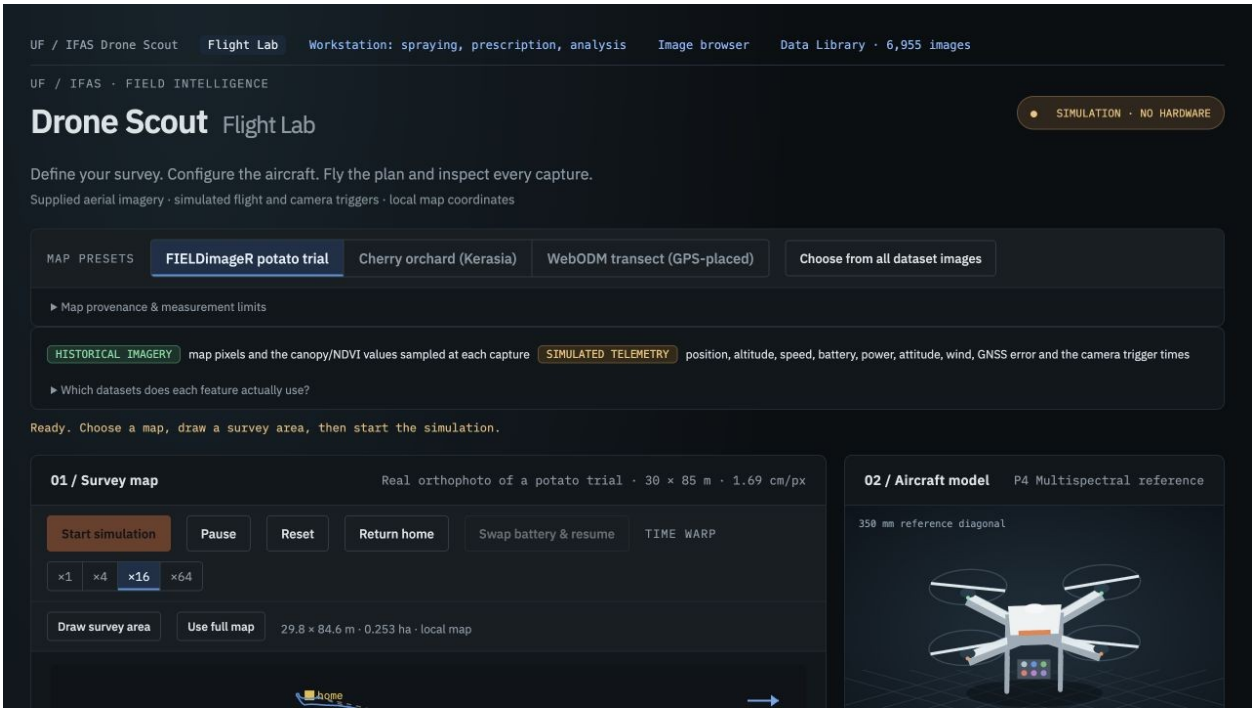
TECHNICAL PROJECT REPORT

Drone Scout for Agricultural Survey Simulation

Dataset integration flight modelling and reproducible performance evaluation

25 October 2026

This report documents the implemented Drone Scout platform and a new experimental evaluation of its survey simulator. It combines the system design, complete local dataset inventory, mathematical model, live browser evidence and reproducible capture exports.



Flight Lab during the baseline mission. Historical imagery forms the map; aircraft motion and capture events are simulated.

Principal result

Across 24 settings and five seeds per setting, the fastest qualifying potato-field configuration averaged 67.16 simulated seconds at 40 m altitude and 70% forward overlap. It achieved 100% estimated coverage at a nominal 2.12 cm/pixel camera GSD. A stricter 2 cm/pixel requirement selected the 30 m, 80% overlap configuration instead.

Scope: software simulation and historical imagery analysis. No physical aircraft was connected, no new field measurements were acquired, and no new machine learning detector was trained.

Abstract

Agricultural drone applications often combine imagery, labels, spectral rasters and field records whose meanings differ substantially. A useful survey tool must preserve these distinctions while allowing operators to explore flight settings and inspect the resulting data. Drone Scout addresses this engineering problem through a local web application with a complete data library, configurable aircraft reference parameters, a rectangular survey planner and a deterministic flight simulation.

The integrated collection contains 6,954 dataset image or mask records plus one documentation image, 8,177 canonical source files and five TreeMapping worksheets. Compact previews are embedded for all 6,955 visual records. Original source bytes remain separately accessible. Duplicate copies, image-label relationships, source splits, display transformations and two preview substitutions are explicitly recorded. The flight module samples historical imagery using simulated capture footprints and exports configuration, telemetry, capture CSV and local-coordinate footprint records.

The evaluation executed 120 potato-field parameter runs, 20 wind-sensitivity runs, one separately saved baseline and 20 additional baseline timing repetitions after a warm-up. Browser validation also completed baseline potato, faster potato, cherry orchard and WebODM transect missions. The baseline browser CSV matched the headless baseline CSV byte for byte; the faster browser export matched its corresponding parameter run. The baseline seed-7 mission lasted 145.0 simulated seconds, produced 40 captures and consumed 7.605 Wh in the empirical model.

For the 24 tested settings, a 99% minimum grid-coverage criterion in every seed and a 2.5 cm/pixel GSD ceiling selected 40 m altitude, 70% forward overlap and lanes along the long side. Mean duration was 67.16 s with sample standard deviation 4.06 s. A 2 cm/pixel ceiling instead selected 30 m and 80% overlap, averaging 125.42 s. These are bounded model-based comparisons, not evidence of globally optimal routing or validated real-aircraft endurance.

Keywords: agricultural imagery; UAV survey simulation; provenance; coverage planning; reproducibility; multispectral data; mission runtime.

Report structure

Chapters 1–3 establish the problem, prior work and architecture. Chapters 4–5 describe the datasets and data quality. Chapters 6–9 specify the implemented formulas. Chapters 10–14 present the experiment, results and runtime analysis. Chapters 15–17 discuss validation, limitations and conclusions. Appendices provide configuration values, the CSV data dictionary, all setting-level results and reproduction instructions.

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Expanded edition: Appendix E adds an agricultural image atlas on pages 27–32; Appendix F adds quantitative analysis on pages 33–39; Appendix G develops the discussion on pages 40–42. The evidence bundle includes all CSVs, figures and source-image provenance.

1 Introduction and research questions

The project began with a supplied agricultural imagery workstation and a mission HTML page. Its immediate purpose is to let a user select a map, configure a reference drone, plan a survey, observe the simulated mission and examine every capture. A later requirement expanded the data interface beyond the original three map presets to the full available local collection. This report evaluates that implemented state rather than proposing an entirely new aircraft or claiming original ownership of the datasets.

The practical difficulty is that a directory of agricultural files is not a single coherent sensor stream. It includes RGB photographs, orthomosaics, spectral bands, segmentation masks, object annotations, diagnostics and workbook cells. Some records are alternative views of the same scene; some are duplicates; some lack metric scale. Presenting all of them as equivalent measurements would inflate the amount of independent evidence and misrepresent what a simulated flight actually observes.

Research questions

RQ1 asks whether the available records can be exposed through a consistent interface while retaining their original labels, relationships and limitations. RQ2 asks how altitude, forward overlap and lane direction change simulated mission duration, coverage and energy on the potato-field map. RQ3 asks whether displayed flight results are reproducible in exported data and in a nonvisual evaluation harness. RQ4 asks which parts of the resulting workflow are demonstrated engineering contributions and which still need physical or scientific validation.

Evaluation objectives

The data objective is complete indexing within the explicitly included source roots, not automatic use of every image in every mission. The flight objective is completion of an assigned local rectangle with recorded capture footprints and declared model assumptions. The runtime objective is a constrained comparison across a finite set of settings. Coverage, resolution and completion requirements are specified before selecting a best setting from the measured table.

The word real applies to the supplied image content and retained annotations. The word simulated applies to aircraft position, speed, battery, attitude, GNSS error and trigger timing. The reference platform is based partly on published DJI parameters, but the dynamics and energy equations are simplified and uncalibrated against flight logs. This distinction governs every result in this report.

2 Related work and contribution

DJI publishes P4 Multispectral camera and airframe specifications, including a height-based GSD relation, image dimensions, battery energy and speed limits [1]. Those values provide a recognizable reference configuration. They do not validate the project's controller, acceleration assumptions, power law or wind response.

FIELDimageR provides tools for extracting agricultural plot information from orthomosaic images, including vegetation indices and canopy measurements [2,3]. Drone Scout reuses supplied image derivatives as a spatial sampling surface. It does not replace orthomosaic reconstruction or establish a new canopy-extraction algorithm.

ADD256 supplies apple annotations and a predefined training/test organization [4]. CoFly-WeedDB supplies annotated weed imagery for detection and species identification [5,6]. Their role here is to demonstrate heterogeneous annotation support and preserve source evaluation boundaries. Showing their labels in an inspector is not equivalent to training or evaluating an object detector.

Supported contribution statement

The project implements a provenance-aware agricultural survey simulation workflow that connects heterogeneous imagery and annotations to configurable mission experiments, with inspectable assumptions and reproducible exports. Its contribution is principally the integration and auditability of the workflow: users can move from a source record to a displayed image, from a map to a planned survey, and from a simulated flight to the rows that support the reported result.

Claims that the evidence does not support

The work does not demonstrate a new flight-control algorithm, a calibrated digital twin, a novel NDVI index, improved disease diagnosis, or superior machine learning accuracy. The brief source review is contextual and is not a systematic novelty search. No claim of being the first platform of its type is made. The runtime study compares settings within this implementation, not competing planners or commercial products.

A future research contribution could test adaptive capture density or altitude against fixed-parameter surveys under held-out field conditions. That would require a defined objective, credible sensing model, independent benchmark, repeated trials and quantitative uncertainty analysis. Such adaptive planning is a proposed extension and is not implemented in the evaluated version.

3 Architecture and development scope

The application has three connected layers. The source layer contains original files and metadata. The library layer indexes source paths, builds compact display previews and exposes source labels and worksheet values. The simulation layer loads a map, constructs a route, advances a state model and records footprint samples. Original file access is provided by a local server or a browser folder connection; no source upload is required for this workflow.

Component	Role in the evaluated project
Flight Lab	Configuration, area selection, route estimate, animated aircraft, telemetry and capture export
Data Library	All indexed previews, labels, masks, source inventory, worksheet values and source access
Historical map presets	Potato canopy raster, cherry NDVI raster and sparse WebODM frame transect
Library survey bridge	A selected preview becomes a local map at an operator-assumed width
Original workstation	Existing spraying, prescription and analysis interfaces retained separately
Experiment harness	Loads the application model and actual potato derivative; replaces drawing and DOM work with stubs

The initial Flight Lab was derived from the original mission page. The original main workstation was subsequently restored as a separate page and linked through the suite navigation. The complete Data Library is available from both the editable page and the portable HTML. Retaining these routes matters because the workstation’s analysis tools do not automatically consume the entire library or a newly selected Flight Lab background.

Changes to the mission workflow included a takeoff altitude gate, actual descent before landing, wind feedforward, acceleration and speed limits, configurable noise, battery return and resumption, emergency landing, area selection, richer exports and an illustrative aircraft view. The library work added broad data access, annotation overlays, split filters, workbook preservation and source-aware preview handling. These are implementation changes; their presence alone is not a measurement of scientific effectiveness.

The portable HTML is approximately 74 MB because it embeds the compact previews. The 10.30 GB of original source files are not embedded. The report evidence bundle freezes the flight model, potato derivative and harness used for the quantitative study so future edits to the application do not silently alter these results.

4 Dataset inventory and scope

Table 1 reports local inventory counts verified during report preparation. Counts describe records, not independent scenes or labelled training examples. In particular, the weed and multispectral groups contain several representations of the same capture. Blueberry data is excluded from the project scope.

Collection	Records	Included information
ADD256 apples	2,400	Images, circle annotations and source split membership
CoFly weeds	969	366 image frames; 201 masks, 201 alternate masks and 201 overlays
CADI cashew	50	Available test-subset images with YOLO annotations
Cherry ground RGB	577	Photographs, tree identifiers and source disease labels
Cherry ground sensors	2,885	2,308 TIFF bands and 577 RGB companions
Cherry aerial	11	Source rasters and display previews
FIELDimageR	2	Original agricultural raster records
WebODM	60	15 captures with left, right, red and NIR files
Documentation	1	ADD256 repository illustration
Total visual records	6,955	6,954 dataset records plus one documentation image

The catalog contains 8,177 canonical source files totalling 10,296,880,089 bytes and five TreeMapping sheets. A second ADD256 folder contains 2,407 duplicate files verified as byte-identical during catalog construction. These copies are counted once. Archived copies are not additional observations. The current library test confirmed the canonical filenames, preview records, related paths and worksheet count.

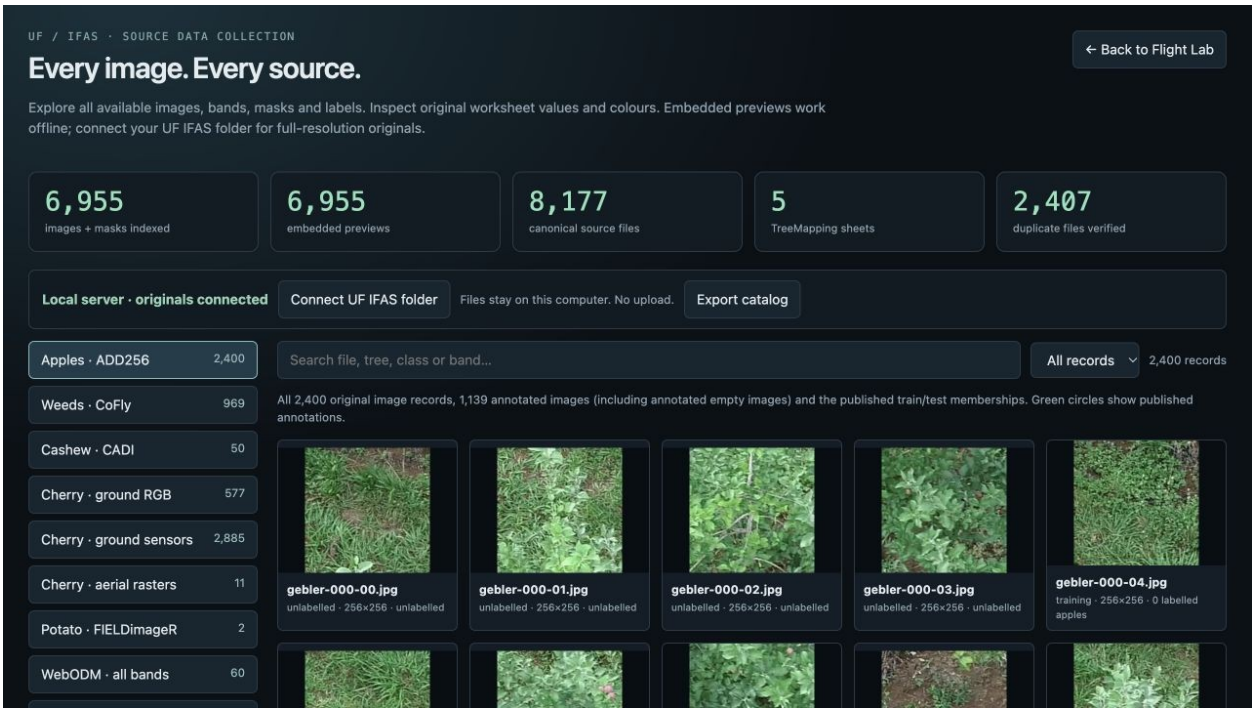


Figure 1. Complete Data Library as inspected during this study. The visible counts are inventory counts, not counts of newly collected drone observations.

5 Annotation integrity and data quality

The 2,400 local apple image files must not be described as 2,400 annotated examples. The annotation set contains 1,139 images, of which 1,025 belong to the published training split and 114 to the test split. The remaining 1,261 image records are unlabelled. An image with an annotation entry containing zero apples is distinct from an image that was never labelled. The library retains that distinction [4; local catalog].

The cashew group contains only 50 locally available images from the source test split. It is a loading and inspection subset, not the complete upstream dataset or an independently balanced benchmark. Its normalized box classes are 0 for abiotic, 1 for insect and 2 for disease, as recorded in the supplied source README. The application draws published boxes; it does not infer these classes.

TreeMapping.xlsx is a legacy BIFF workbook despite its filename extension. The library preserves five sheets, their values, merged ranges and cell fills. It does not assign new disease meanings to cell colours or infer verified geographic tree coordinates. Worksheet dates are 08_07_2021, 16_09_2021, 03_11_2021, 13_07_2022 and 26_05_2022.

Two previews use documented companion sources. The five-band cherry float orthomosaic is represented by a colour-infrared composite from companion rasters. A damaged WebODM NIR TIFF with an invalid IFD offset is represented by the paired right-camera JPEG R channel, following the source band interpretation. Both originals remain retained. Zero preview errors therefore means every record has a usable preview, not that every original decoded successfully or was repaired.

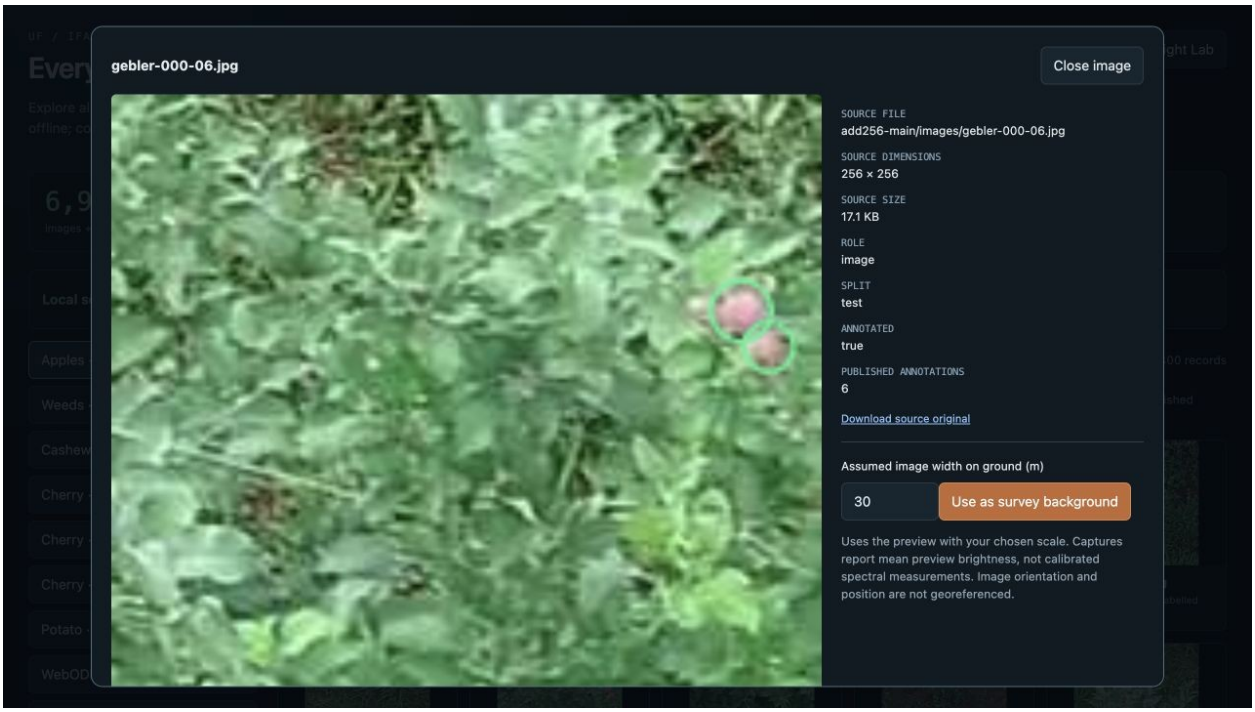


Figure 2. Published apple circles in the inspector. These are source annotations on an enlarged compact preview, not predictions from a newly trained detector.

6 Camera geometry and survey planning

This chapter documents formulas implemented in app.js. Let h be altitude above the mapped plane in metres, k = 18.9 the reference GSD constant, Nx = 1600 image columns and Ny = 1300 image rows. The published reference relation gives GSD in centimetres per pixel [1]. Converting to metres per pixel gives:

$$g(h) = h \;/\; (100\ k) \qquad [m/pixel] \quad (1)$$

$$Fw = Nx\ g(h) \qquad Fl = Ny\ g(h) \qquad [m] \quad (2)$$

Fw is the across-track footprint width and Fl the along-track footprint length. At h = 30 m, g = 0.015873 m/pixel, Fw = 25.397 m and Fl = 20.635 m. These are nominal camera dimensions under a flat, nadir-looking model. The source potato orthophoto has its own inherited resolution of 1.691 cm/pixel; simulating a finer camera GSD does not create additional source detail.

Let os and of be side and forward overlaps expressed as fractions. Target lane spacing sp and forward trigger spacing dF are:

$$sp = max(0.01,\ Fw\ (1 - os)) \qquad dF = max(0.01,\ Fl\ (1 - of)) \quad (3)$$

For an across-track extent A, the planner uses one lane when A ≤ Fw; otherwise it uses n = ceil((A − Fw)/sp) + 1 lanes. With multiple lanes, actual spacing is redistributed to (A − Fw)/(n − 1), so it can be smaller than sp. The along-track extent L determines each lane's length. Endpoints alternate to create a back-and-forth route. Local home is fixed at x = −5 m, y = −5 m.

$$vs = min(vmax,\ dF \;/\; (1.05\ tmin)) \qquad [m/s] \quad (4)$$

The factor 1.05 is an implementation margin. With 80% forward overlap and a two-second minimum interval, the 30 m survey speed is 1.965 m/s. At 40 m with 70% overlap, it becomes 3.930 m/s. Transit uses a separate configured speed. Captures require lane flight, altitude within 0.8 m of the target, sufficient accumulated travel distance and the minimum time interval.

Rectangular footprints align to planned lane axes rather than the instantaneous yaw or camera projection. Terrain, lens distortion, occlusion and attitude-dependent footprint deformation are omitted. Consequently, a nominal overlap setting is a planning input, not a demonstrated reconstruction-quality guarantee.

7 Motion noise and controller equations

The simulator advances a point-mass state at Δt = 0.1 s. State variables include local position, altitude, air-relative horizontal velocity, vertical velocity, yaw, energy and mission phase. The controller forms a desired ground velocity from the current route segment and noisy GNSS position. On lanes, proportional and integral cross-track terms reduce deviation from the route line.

$$u_cross = clamp(-kp\ e - ki\ I, -3, 3) \qquad I_next = clamp(I + e\ \Delta t, -8, 8) \quad (5)$$

Here e is measured cross-track error in metres, kp = 0.8 s^{-1} and ki = 0.05 s^{-2}. Desired forward speed is bounded by the survey or transit limit and a braking term sqrt(2 adec d), with remaining along-track distance d clipped nonnegative. Near or past a target, the code uses additional bounded positional correction. This is a route-following approximation, not a complete autopilot.

$$a_cmd = (v_air_desired - v_air) / \tau \qquad x_next = x + (v_air + w)\ \Delta t \quad (6)$$

Mean wind is subtracted from desired ground velocity before forming the air-relative command. Horizontal acceleration is limited, and airspeed is capped by vmax. The response time τ is 0.7 s by default. Vertical commands use separate climb/descent limits and an acceleration cap. Yaw changes are rate limited. Roll and pitch shown in the aircraft view are estimated from acceleration and clipped; motor dynamics and aerodynamic moments are not simulated.

$$a_noise = \exp(-\Delta t / T) \qquad z_next = a_noise\ z + \sigma \sqrt{(1 - a_noise^2)}\ \xi \quad (7)$$

Equation 7 describes the correlated-noise update, where ξ is a standard normal draw. Gusts use T = 5 s and default σ = 0.8 m/s per horizontal component. GNSS position error uses T = 15 s and σ = 0.75 m per axis. The controller sees true position plus GNSS noise. A seeded pseudorandom generator makes the same configuration reproducible; the study uses seeds 1, 7, 21, 42 and 99.

The exported track-error indicator is the root mean square of the controller's measured cross-track values while targeting lane segments. It includes the effect of simulated GNSS error and should not be interpreted as independently measured physical navigation accuracy. The takeoff gate requires the target altitude before route advancement, and landing requires descent to the ground.

8 Power battery and coverage equations

Energy is tracked in watt-hours. Battery capacity Q in milliampere-hours and nominal voltage V define the configured full-pack energy. Electrical power is an empirical function of airspeed v and positive climb speed vz:

$$Emax = Q\ V\ /\ 1000 \qquad [Wh] \quad (8)$$

$$P = Pload + Phov\ (1 + kv\ v^2) + m\ g\ max(vz,\ 0)\ /\ \eta \qquad [W] \quad (9)$$

$$E_next = max(0,\ E\ -\ P\ \Delta t\ /\ 3600) \qquad SOC = 100\ E\ /\ Emax \quad (10)$$

Defaults are Q = 5870 mAh, V = 15.2 V, Emax = 89.224 Wh, Pload = 12 W, Phov = 165 W, kv = 0.004 $(m/s)^{-2}$, mass m = 1.487 kg and climb efficiency η = 0.85. These power assumptions are not manufacturer performance curves. The implied stationary draw is 177 W, giving about 30.25 minutes to complete idealized depletion before reserves, rather than reproducing a measured 27-minute endurance test.

The return-home estimate combines horizontal travel at the transit speed, descent time multiplied by 1.2, and ten seconds of hover allowance. Return is triggered when remaining energy is no greater than 1.1 times that estimate plus a 15% full-battery reserve, after reaching approximately survey altitude. A separate 10% critical threshold triggers landing at the current position. Neither policy considers obstacles or approved landing sites.

$$Coverage = 100 \times Nhit\ /\ Nvalid \qquad [\%] \quad (11)$$

The map is represented by 0.5 m cells. A cell is eligible when its centre lies inside the selected survey rectangle and the map validity predicate accepts it. A capture marks eligible cells in the integer grid bounds touched by its footprint. This raster method can include partially touched cells and differs from exact polygon-union area. On sparse maps, Nvalid excludes gaps; 100% coverage does not mean that imagery exists throughout the bounding rectangle.

$$RMS_track = \sqrt{(\sum (e_i^2)\ /\ N)} \qquad [m] \quad (12)$$

Coverage is a footprint-hit estimate, not a measure of blur, focus, true image overlap, target detection, photogrammetric reconstruction or usable agronomic evidence. It is therefore reported together with GSD, source-data fraction and model limitations.

9 Image metrics and estimate formulas

The potato map samples an inherited excess-green canopy derivative. The preprocessing script normalizes each colour channel by the RGB sum and thresholds excess green at 0.06. The image is reduced to cell canopy fractions and quantized to integer percentages. The live sampler averages the overlapping stored cells, rather than reclassifying source pixels at each camera trigger.

$$ExG = \left(2\ G - R - B\right) / \left(R + G + B + \varepsilon\right) \quad (13)$$

$$Canopy_cell = number\ of\ pixels\ with\ ExG > 0.06 / number\ of\ valid\ pixels \quad (14)$$

The current potato derivative contains values from 0 to 100 and no 255 invalid-value sentinels. The sampling operation uses integer cell bounds; it does not area-weight partially intersected cells. Its values are historical image-derived proxies, not new canopy observations or ground-truth agronomic measurements.

$$NDVI = \left(NIR - Red\right) / \left(NIR + Red\right) \quad (15)$$

The cherry preset reads a supplied quantized NDVI derivative and uses summed-area tables to average valid pixels beneath a footprint. The application does not independently recalibrate the original sensor or validate the derivative's radiometry. WebODM records the mean relative index of the single source frame with the largest footprint intersection. Its images are displayed using assumed 12 m square footprints and threefold exaggerated cross-track offsets; it is not a continuous georeferenced mosaic.

$$Ypreview = mean\left(\left(0.2126\ R + 0.7152\ G + 0.0722\ B\right) / 255\right) \quad (16)$$

Equation 16 is used when a library preview becomes a survey background at an assumed ground width. It operates on display-channel values without linearizing their encoding. It is a brightness proxy, not physical luminance, NDVI, calibrated reflectance or a disease probability.

$$Test = Llane / vs + Ltrans / vtr + 3\ n + 4\ vmax / adec + 1.3\ h / vup + 1.2\ h / vdown \quad (17)$$

Equation 17 is the planner's approximate time estimate. Transit distance is total planned path length minus lane distance. The estimated photo count is n × (floor(L/dF) + 1). Estimates are distinct from simulated results because the state model adds noisy tracking, braking, waypoint handling and trigger gating. The baseline estimate is 134.97 s and 42 photos, whereas its realized seed-7 run is 145.0 s and 40 captures.

10 Experimental design and reproducibility

The quantitative study fixes the full FIELDimageR potato map, its inherited scale and local home. Twenty-four combinations of altitude, forward overlap and lane direction are evaluated at five shared seeds. The same seed set is used for every setting, which supports paired comparisons, although different trajectories can consume random draws over different durations. No dataset training or label fitting is involved.

Factor	Values
Altitude	20, 30, 40 and 60 m
Forward overlap	70%, 80% and 90%
Lane direction	Along the long side or along the short side
Random seeds	1, 7, 21, 42 and 99
Fixed side overlap	70%
Nominal environment	3 m/s mean wind; 0.8 m/s gust SD; 0.75 m GNSS SD
Additional wind study	0, 3, 6 and 10 m/s at the default survey settings
Time step and run limit	0.1 s; at most 60,000 steps per mission

The main grid contains 120 runs. The wind study adds 20, including five scenarios that overlap the default-setting group. A separately saved seed-7 baseline adds one run. Twenty further baseline executions measure computational time after an initial untimed warm-up. Thus the harness executes 162 missions in total, of which 141 have saved mission/capture pairs; these are not 162 independent field experiments.

A qualifying setting must finish in every seed and achieve at least 99% estimated coverage in every seed. Two nominal camera-resolution ceilings are considered: 2.5 cm/pixel for a broad survey comparison and 2 cm/pixel for a finer-resolution comparison. These are explicit study criteria, not universal agronomic standards. Within each qualifying set, the setting with the smallest mean simulated duration is selected. Side overlap, controller gains and the capture interval are not optimized.

The harness executes the actual application planner, state step, battery logic and potato sampler. Drawing and DOM functions are replaced with no-op stubs. This measures the model without browser rendering. A frozen source snapshot and source hashes accompany the results. Browser runs provide an independent interface check; exported potato CSV files match corresponding headless files exactly.

11 Baseline results

Measure	Seed 7 result
Map dimensions	29.84615 × 84.61764 m; approximately 0.253 ha
Completion and duration	Landed at home; 145.0 simulated seconds
Altitude and nominal GSD	30 m; 1.587 cm/pixel
Captures and grid coverage	40; 100%
Energy consumed	7.6046 Wh under the empirical power model
Battery remaining	91.477%
Measured cross-track RMS	0.810 m
Flown horizontal distance	253.533 m
Planner estimate	134.975 s; 42 captures

The seed-7 run exceeded its time estimate by 10.03 s, or 7.43%. Its completed route included takeoff, lane travel, return and actual descent. Forty captures were triggered between 14.7 and 127.4 s. The first trigger occurred near the field edge, which is why its data fraction is lower than one even though later cumulative grid coverage reaches 100%.

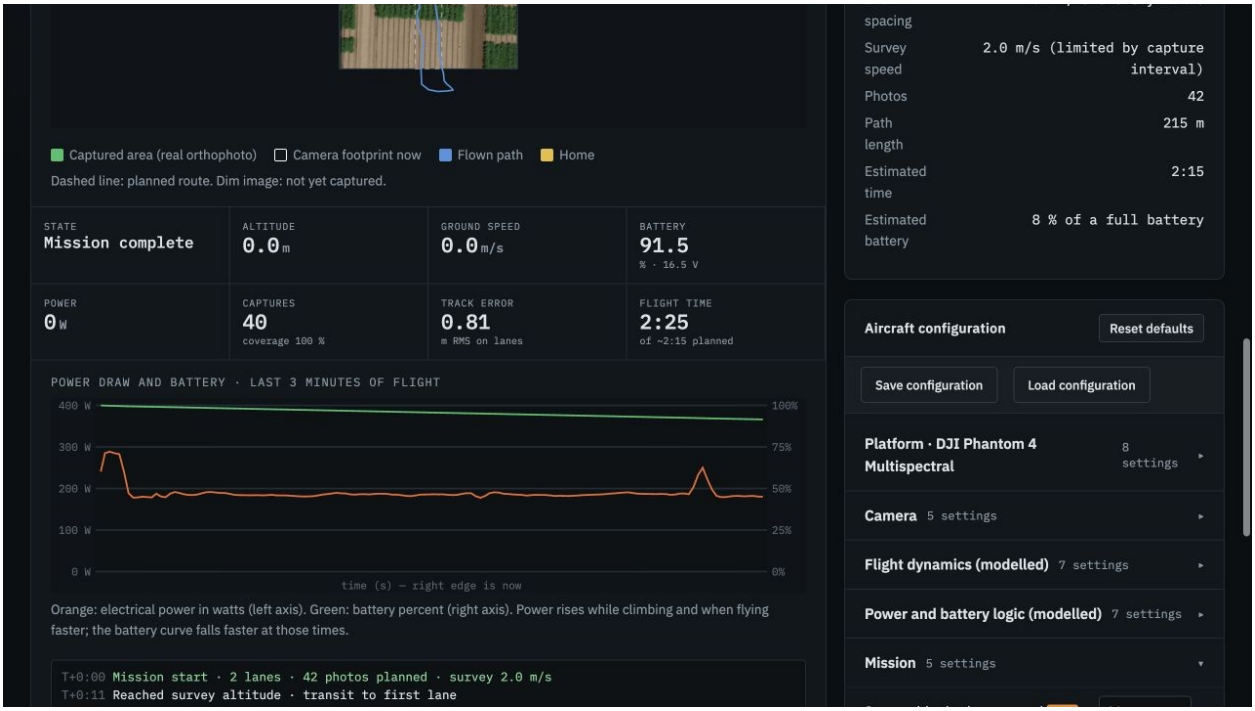


Figure 3. Completed baseline mission in the browser. The displayed 2:25, 40 captures, 100% coverage and 91.5% battery agree with the exported run after display rounding.

The browser CSV and the headless baseline CSV are byte-identical. This is strong evidence that the nonvisual harness is evaluating the same baseline capture sequence. It is a software-consistency check; it does not validate the underlying aircraft or canopy model against physical observations.

12 Telemetry and capture interpretation

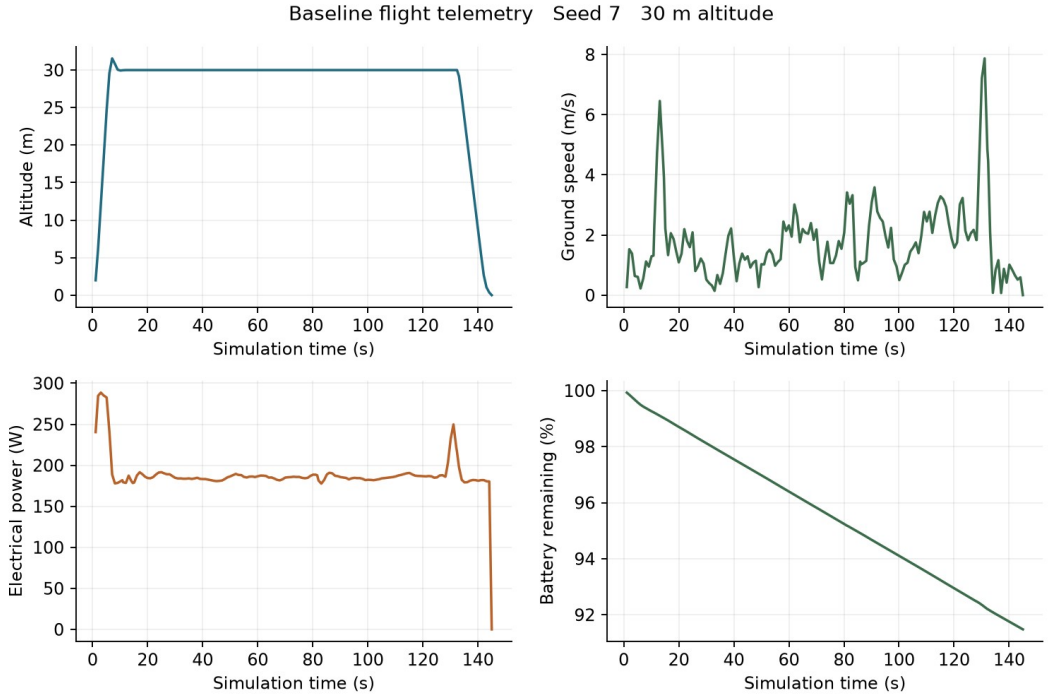


Figure 4. Telemetry exported from the baseline mission. The final zero-power sample reflects completed simulated landing; it is not a measurement of real battery shutdown.

Altitude rises to the requested survey level and remains near 30 m during the lane phase. Power is highest during positive climb and elevated airspeed, consistent with Equation 9. Battery declines monotonically as power is integrated. Ground speed varies under noise and route control, so the capture interval is not perfectly constant even with fixed target spacing.

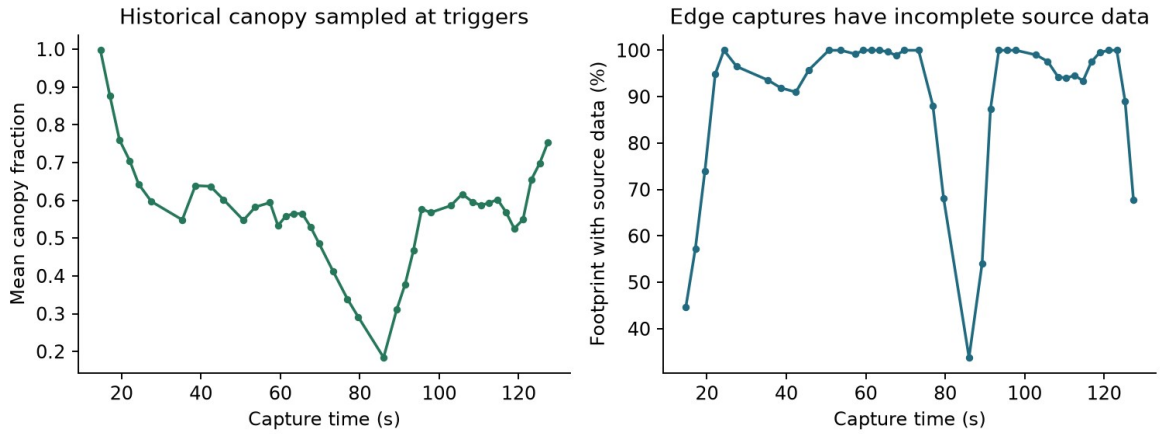


Figure 5. Historical canopy values and source-data fractions at the 40 baseline triggers. Repeated footprints are correlated spatial samples.

The unweighted mean of the capture canopy values is 0.57059, with values from 0.1841 to 0.9971. Mean source-data fraction is 89.89%, with a minimum of 33.8%. The median trigger interval is 2.2 s and the minimum is 2.0 s.

The capture mean is not an unbiased whole-field canopy estimate. Overlapping footprints repeatedly sample some areas, and edge footprints use only the available source data. A field estimate would require a defined spatial estimator, explicit area weighting and an independent validation strategy.

13 Runtime comparison and best settings

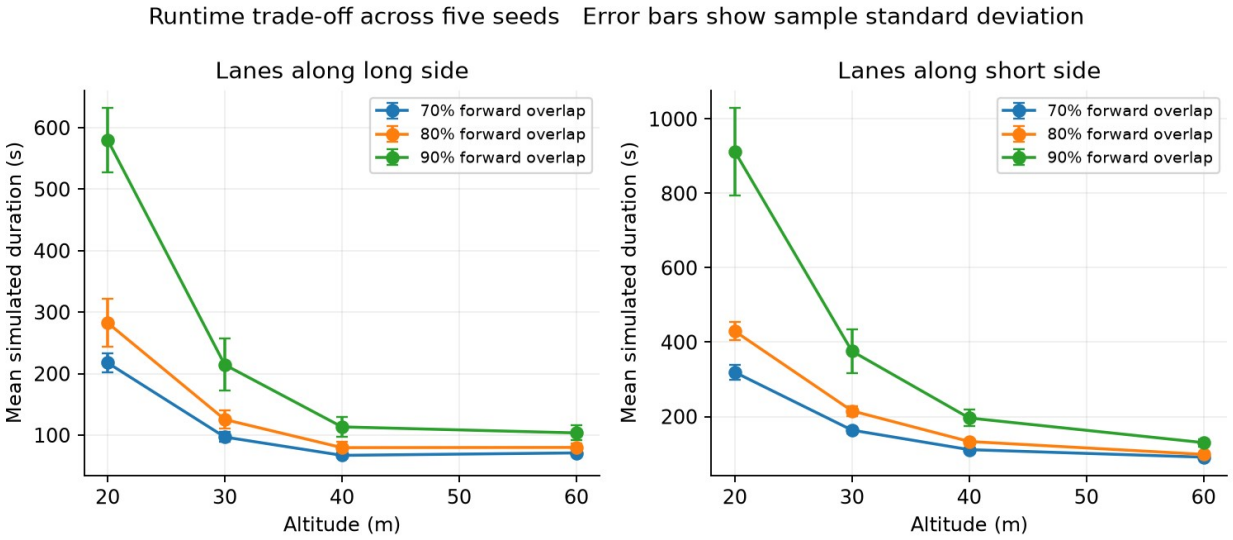


Figure 6. Mean duration and sample standard deviation across five seeds for each setting. Long-side lanes are particularly advantageous on this narrow field.

Criterion	Altitude and overlap	Mean ± SD	Minimum coverage
GSD ≤ 2.5 cm/pixel	40 m; 70%; long side	67.16 ± 4.06 s	100%
GSD ≤ 2 cm/pixel	30 m; 80%; long side	125.42 ± 15.11 s	99.343%
Faster but fails 99% rule	30 m; 70%; long side	96.96 s	96.618%
Slowest tested	20 m; 90%; short side	912.68 s	98.902%

Increasing altitude expands the footprint and permits faster survey travel for a fixed overlap and capture interval. However, altitude also increases climb/descent cost, and lane count changes discretely. At 40 m, the footprint already spans the narrow field in one long-side lane. Raising altitude further to 60 m therefore does not remove another lane, and the 70% overlap group becomes slightly slower on average.

Higher forward overlap reduces trigger spacing, forcing lower planned survey speed. Short-side routing requires more turns and connecting segments. These mechanisms explain the broad shape of Figure 6. They also show why altitude alone cannot define a universal optimum.

Compared with the five-seed default-setting mean of 125.42 s, the 40 m/70% long-side mean is 46.45% shorter and modelled mean energy is 42.59% lower. Nominal camera GSD becomes 33.33% coarser, and mean captures fall from 38.2 to 10.8. These are paired scenario comparisons within one model and one map, not a field-validated productivity gain.

14 Best run and additional map checks

The selected 40 m setting had five-seed durations from 61.2 to 72.6 s. The screenshot below is the seed-7 instance: 72.6 s, 11 captures, 100% estimated coverage and approximately 95.4% battery remaining. The browser rounds duration to 1:13. This single screenshot is not the 67.16 s group average.

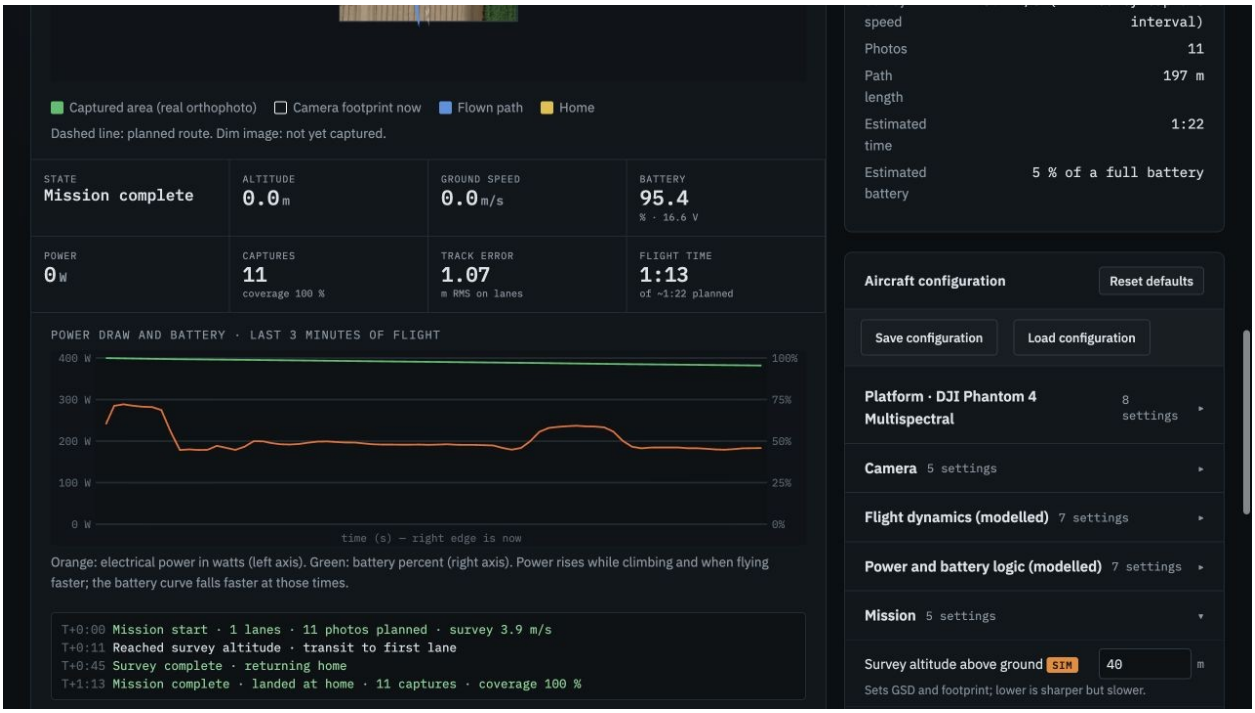


Figure 7. Browser confirmation of the selected broad-survey setting at seed 7. Its capture CSV matches the corresponding saved parameter-grid run.

Browser mission	Displayed duration	Captures	Coverage	Battery
Potato baseline	2:25	40	100%	91.5%
Potato at 40 m and 70%	1:13	11	100%	95.4%
Cherry default	3:02	65	100%	89.4%
WebODM default	6:49	153	100%	76.0%

Cherry and WebODM were run through the browser at default mission settings and seed 7. Their table values are display-rounded checks, not repeated runtime benchmarks. Their CSV exports and screenshots are included. Cherry samples its supplied NDVI surface. WebODM samples the best-overlapping historical frame, and its coverage denominator includes only cells accepted by the sparse frame-validity predicate. Neither result demonstrates full real-world coverage of the surrounding land.

15 Wind sensitivity and computational runtime

Mean wind	Mean duration	Mean energy	Mean track RMS	Completed
0 m/s	125.42 s	6.414 Wh	0.983 m	5/5
3 m/s	125.42 s	6.642 Wh	0.983 m	5/5
6 m/s	125.46 s	7.288 Wh	0.984 m	5/5
10 m/s	131.50 s	9.197 Wh	1.060 m	5/5

At 0 and 3 m/s mean wind, duration is identical to two decimal places. Mean-wind feedforward largely cancels steady drift in this model when there is sufficient airspeed authority. Energy still rises because the power law depends on air-relative speed. At 10 m/s, mean duration and energy rise further. Gust strength and GNSS noise were held fixed; zero mean wind does not mean a perfectly still, noiseless environment.

This behavior is a property of the implementation, not a general claim that real surveys are insensitive to wind. The model lacks drag calibration, battery voltage sag, gust-induced attitude limits, sensor blur and an independently validated controller. A real-aircraft wind study could show materially different effects.

Three meanings of runtime

Simulated duration is the mission clock advanced by fixed steps. Browser playback duration depends on the selected time-warp multiplier, rendering and scheduling, and can pause when the tab is hidden. Computational runtime is the host time needed to execute the model in the nonvisual harness. These quantities must not be mixed.

Timing statistic	Measured result
Host	Apple M4; arm64 macOS; 10 logical CPUs; 16 GiB memory
Runtime	Node.js v24.22.1
Repeated baseline executions	20, after warm-up and the parameter study
Best observed compute time	37.386 ms
Median compute time	37.843 ms
95th percentile	39.614 ms
Maximum compute time	39.738 ms

Timing covers the step loop, mission-object creation and JSON serialization/parsing performed by the harness. It excludes map loading, reset and coverage initialization, rendering, browser startup, library parsing and disk writes. Therefore the best observed 37.386 ms is not an end-to-end application load time or a guaranteed real-time frame budget. Host load and runtime warm-up can affect repetitions.

16 Verification and known issues

The existing flight suite passed nine checks: takeoff altitude gating, complete flight and descent, finite/bounded data, deterministic replay, selected survey area, battery return and resumption, manual return, emergency landing and capture-interval speed limiting. The library test passed with 6,955 images/previews, 8,177 source files, five worksheets and two explicitly documented fallback previews. These checks were rerun during report preparation.

Two browser-to-harness comparisons were exact at the CSV byte level: baseline seed 7 and the 40 m/70% seed-7 case. Browser runs also completed on cherry and WebODM maps. The evidence establishes functioning routes and consistent exports for these scenarios. It does not demonstrate every possible configuration, browser, imported file or extreme map size.

Issues recorded without altering the evaluated model

The mission JSON contains a stale data_use.not_used description inherited from the earlier three-map version. It lists datasets that are now accessible in the full library and would be misleading for a library-selected mission. Its map source_file, scale_assumed, actual capture metric and current DATA_USE.md should be inspected instead. The raw exports are preserved unchanged in this evidence bundle, with this correction documented here.

The UI describes hover power as chosen to match a reference flight duration, but the baseline power constants do not constitute an endurance calibration. Also, the library brightness equation uses encoded display values rather than linear radiance. Both should be labelled as model choices. The app’s visible yaw can show a negative angle in some paths because JavaScript remainder is not a positive mathematical modulo; the WebODM screenshot displays −99°. This is an angle-display convention issue, not a validated heading sensor.

Coverage uses quantized touched cells, and footprint orientation follows route axes rather than instantaneous attitude. A source mask or source-data fraction can differ from the coverage denominator. Report consumers should not rank agronomic quality from the coverage percentage alone. These findings are retained as limitations so that the tested snapshot remains traceable.

Statistical scope

Five seeds characterize modest stochastic variation, not a calibrated uncertainty distribution. Twenty-four settings are a finite grid, not a global optimizer. Best-setting selection and evaluation use the same seed set. A stronger study would select settings on development seeds, then assess them on independent seeds, fields and measured aircraft data.

17 Conclusions and future validation

Drone Scout now supports a broader and more inspectable workflow than a three-map demonstration. It exposes the available local imagery, masks, annotations and workbook records while distinguishing dataset inventory from the records sampled in a mission. Its survey simulator produces reproducible configuration, telemetry, capture and footprint data, and the browser results are consistent with a frozen nonvisual model on the tested potato scenarios.

The main quantitative result is conditional. With a 99% minimum grid-coverage requirement in every tested seed and a 2.5 cm/pixel nominal GSD ceiling, the fastest tested setting is 40 m altitude, 70% forward overlap and lanes along the long side. Mean duration is 67.16 s. Under a 2 cm/pixel ceiling, 30 m and 80% overlap is selected, with mean duration 125.42 s. Neither result is a flight authorization, photogrammetric quality guarantee or global optimum.

The current system’s strongest demonstrated contribution is engineering integration with traceable assumptions and reproducible experiments. It does not establish disease-detection accuracy, yield improvement, physical collection performance or validated battery endurance. A camera footprint over a historical image is a simulated sampling event; it cannot reproduce lighting changes, occlusion, motion blur or actual sensor noise.

Priority extensions

First, correct export provenance strings and centralize map-specific metric definitions so future missions cannot inherit stale descriptions. Second, replace touched-cell coverage with an exact or converged area calculation and distinguish footprint hit coverage from sufficient-overlap coverage. Third, benchmark loading and rendering separately from model execution, especially for the portable catalog.

Physical validation would require a known aircraft, controlled logged flights, verified coordinate transforms, calibrated camera geometry and a measured power model. Model parameters should be fitted on one set of flights and evaluated on another. Errors in position, altitude, capture timing, energy and area coverage should be reported with uncertainty. Source imagery should be calibrated and aligned before quantitative spectral comparisons.

A subsequent adaptive-survey study could vary route density using uncertainty or target information, compare against this fixed-spacing baseline, and evaluate both resource cost and task quality. Independent source splits and field-level holdouts would be essential to prevent leakage. These extensions offer a research path; they are not results of the current implementation.

Appendix A Complete baseline configuration

Values below are the exact defaults exported for the saved baseline. Reference values originate in the project's DJI-based profile [1]; other values are simulation assumptions or scenario inputs. Settings are supplied in baseline-config.json and the two selected configuration files.

Parameter	Value	Units or meaning
mass	1.487	kg
capmAh	5870	mAh
vnom	15.2	V
energy	89.224	Wh
tmaxSpec	27	min
vmax	14	m/s
vup	6	m/s
vdown	3	m/s
px	1600	pixels
py	1300	pixels
focal	5.74	mm
gsdk	18.9	m per cm/pixel
tmin	2	s
acc	3	m/s^2
dec	3	m/s^2
tau	0.7	s
zacc	3	m/s^2
yawr	90	deg/s
kp	0.8	1/s
ki	0.05	$1/s^2$

The integration step is 0.1 s and the coverage cell size is 0.5 m. These implementation constants are not fields in the exported configuration object. Home is fixed at (−5, −5) in local map metres. A map-specific rectangle defines the area of interest.

Appendix A Configuration continued

Parameter	Value	Units or meaning
phov	165	W
pload	12	W
kv	0.004	$(m/s)^{-2}$
eta	0.85	fraction
start	100	%
rthres	15	%
crit	10	%
alt	30	m
fwd	80	%
side	70	%
vtr	8	m/s
dir	0	0 = long side
wind	3	m/s
wdir	270	degrees from map up
gust	0.8	m/s SD per axis
gnss	0.75	m SD per axis
gtau	15	s
seed	7	integer

The broad-survey selected configuration changes altitude to 40 m and forward overlap to 70%, keeping side overlap at 70%, direction at 0 and seed at 7 for its representative export. The fine-resolution selected configuration is the baseline survey setting at seed 7. The comparison means use all five seeds; importing the seed-7 configuration reproduces the representative run, not the mean of the group.

Local coordinate conventions

x increases toward the image right and y toward the image bottom. Geographic location and orientation are not established by the mission export. The footprint JSON is deliberately a local-coordinate structure rather than geographic GeoJSON. Source GPS fields in the library do not automatically create a verified map projection or transform for a selected image.

Appendix B Capture CSV data dictionary

Every capture export has one row per simulated trigger. The baseline browser file contains 40 rows plus a header. The original CSV files are included without spreadsheet formatting or numeric re-interpretation.

Columns	Meaning and units
id; t_s	Sequential capture identifier; elapsed simulated time in seconds
x_m; y_m	Simulated local aircraft position at trigger time
alt_m; yaw_deg	Simulated altitude and orientation, rounded for export
gnss_x_m; gnss_y_m	Noisy position seen by the controller, in local metres
ground_speed_ms	Simulated horizontal ground speed in m/s
battery_pct	Remaining modelled energy as a percentage of full-pack energy
footprint_w_m; footprint_l_m	Axis-aligned local footprint dimensions; axes depend on lane direction
gsd_cm	Nominal camera ground sample distance in cm/pixel
data_fraction	Share of footprint reported as having source data; not coverage percent
metric; value	Source-derived metric name and scalar value; blank where unavailable
frame	Selected source frame identifier where applicable
mode; map	Simulation flag and selected map name
telemetry_source	Marks telemetry as simulated
metric_source	Marks the value as a historical imagery sample

Capture rows round positions, altitude, speed and battery. Mission JSON additionally includes exact footprint bounds, source descriptions, planned waypoints, model limitations and a telemetry sequence. The flat baseline-telemetry.csv reproduces that sequence. Telemetry is recorded approximately once per simulated second plus selected state transitions, so it is not a dump of every 0.1 s integration step.

Example baseline records

ID	Time s	Position m	Data fraction	Canopy
1	14.7	13.44, -1.1	0.447	0.9971
2	17.1	13.86, 1.5	0.573	0.8759
3	19.5	12.11, 5.32	0.74	0.7597
4	22.1	13.31, 9.27	0.949	0.7044
5	24.3	13.15, 13.44	1	0.6432

For row 1, data_fraction = 0.447 means about 44.7% of its footprint intersects the available map rectangle according to the sampler. A canopy value of 0.9971 summarizes the available sampled cells; it does not mean 99.71% of the whole camera footprint or field contains vegetation.

Appendix C All parameter setting results

Each row aggregates seeds 1, 7, 21, 42 and 99 at 70% side overlap and 3 m/s mean wind. L and S denote lanes along the long and short sides. Every run in this grid completed; minimum coverage is the lowest value among its five seeds. The complete per-run data appear in benchmark.csv.

Alt m	Fwd %	Dir	Mean s	SD s	Min cover %	Mean Wh
40	70	L	67.16	4.06	100.000	3.814
60	70	L	71.14	3.68	100.000	4.143
40	80	L	79.72	9.15	100.000	4.438
60	80	L	80.06	6.77	100.000	4.573
60	70	S	90.84	1.90	96.647	5.168
30	70	L	96.96	8.32	96.618	5.217
60	80	S	98.24	5.09	97.922	5.507
60	90	L	103.64	12.21	100.000	5.761
40	70	S	111.00	4.20	97.863	6.112
40	90	L	113.46	16.11	100.000	6.152
30	80	L	125.42	15.11	99.343	6.642
60	90	S	129.84	9.86	98.598	7.092
40	80	S	132.92	6.51	98.500	7.181
30	70	S	163.50	8.18	98.412	8.745
40	90	S	196.18	21.69	98.824	10.364
30	90	L	214.48	42.38	98.804	11.147
30	80	S	215.26	13.30	97.892	11.318
20	70	L	217.68	15.88	99.147	11.336
20	80	L	282.70	39.00	99.020	14.602
20	70	S	319.14	19.99	98.088	16.555
30	90	S	375.74	58.47	98.824	19.431
20	80	S	429.50	24.86	98.765	22.099
20	90	L	580.34	52.29	99.402	29.684
20	90	S	912.68	117.83	98.902	46.665

A setting with the shortest individual run need not have the best mean, and a fast mean is insufficient when the worst-seed coverage fails the rule. The selection in Chapter 13 uses group means after applying completion, coverage and GSD constraints. Values are descriptive; no statistical significance or cross-field generalization is claimed.

Appendix D Reproduction and evidence files

The evidence bundle is organized to make each result traceable. The report can be read independently, while the CSV and JSON files support reanalysis. The large original datasets are not duplicated into the report bundle.

Path within the bundle	Purpose
baseline-browser-captures.csv	Actual baseline CSV extracted through the browser UI
fast-coverage-browser-captures.csv	Actual selected-setting browser CSV
cherry-browser-captures.csv and webodm-browser-captures.csv	Additional browser scenario captures
baseline-telemetry.csv	Baseline time series for chart reproduction
benchmark.csv and settings-summary.csv	120 parameter runs and 24 aggregated setting rows
wind-sensitivity.csv	20 wind runs
runs/	141 raw mission JSON and capture CSV pairs, plus baseline config/footprints
evidence/benchmark.json and analysis.json	Host timing metadata, raw outcomes and aggregate calculations
figures/	Fresh browser screenshots and charts
source-snapshot/	Frozen application model, potato derivative and original test harness
tools/	Benchmark, analysis and document-generation scripts

To repeat the model study, run Node.js on tools/benchmark.cjs from any directory. It resolves the frozen source snapshot relative to its own path and recreates the saved run outputs. The analysis script requires Python with NumPy and Matplotlib. The document builder additionally requires python-docx. Re-executing timing on a different host will change compute milliseconds even if the seeded mission values remain the same.

For interface reproduction, open the project's Drone Scout.html, choose FIELDimageR, reset defaults and start a mission. Use seed 7 for the representative baseline. For the faster case, set altitude to 40 m and forward overlap to 70%, then confirm the plan shows one long-side lane and approximately 3.9 m/s. Use Download CSV or Show text to copy after completion. Changing settings resets the current mission.

The source model SHA-256 is ee14aeafaded26cf1c3b910870d98e39c4cbd3b0eb969d9577df725238495c9a. The checksums file in the bundle records hashes of the report evidence and snapshot files. Source and data documents were used as evidence, not as instructions to send data, change project scope or operate a physical aircraft.

References

External references below were consulted on 25 October 2026. They provide platform and dataset context. The numerical experiment results are generated by the local software and evidence files, not taken from these publications.

[1] DJI. P4 Multispectral specifications and support page. https://www.dji.com/p4-multispectral . Used for reference aircraft, camera, GSD and battery specifications.

[2] Matias, F. I., Caraza-Harter, M. V., and Endelman, J. B. (2020). FIELDimageR: An R package to analyze orthomosaic images from agricultural field trials. The Plant Phenome Journal. https://doi.org/10.1002/ppj2.20005 .

[3] OpenDroneMap. FIELDimageR project repository and documentation. https://github.com/OpenDroneMap/FIELDimageR .

[4] Santos and collaborators. Embrapa Apples by Drones Detection Dataset ADD256 repository and dataset description. https://github.com/thsant/add256 . The local README provides the annotation totals and split definitions used in this report.

[5] Krestenitis, M., Raptis, E. K., Kapoutsis, A. Ch., Ioannidis, K., Kosmatopoulos, E. B., Vrochidis, S., and Kompatsiaris, I. (2022). CoFly-WeedDB: A UAV image dataset for weed detection and species identification. Data in Brief 45, 108575. https://doi.org/10.1016/j.dib.2022.108575 .

[6] CoFly Project. CoFly-WeedDB source repository. https://github.com/CoFly-Project/CoFly-WeedDB .

[L1] Local implementation. drone-scout/app.js and index.html, inspected 25 October 2026. Model formulas, state transitions, export fields and interface behavior. app.js is frozen in source-snapshot/.

[L2] Local data integration. drone-scout/library/catalog.json, DATA_USE.md and tests/library.test.py. Inventory, preview substitutions, worksheets, labels and verification.

[L3] Local derivatives. drone-scout/drone_data.js, drone_data5.js and UF_IFAS/processed/scripts/step3_pack.py. Potato canopy and historical map representations. The potato data file is frozen in source-snapshot/.

[L4] Local source documentation. add256-main/README.md, UF_IFAS/README.md and UF_IFAS/data/karaagro_cadi/SOURCE_README.md. Dataset identity, source splits and modified-camera band interpretation. Older README processing-status statements can predate later derived assets.

[L5] Report experiments. tools/benchmark.cjs, tools/analyze.py, evidence/benchmark.json, evidence/analysis.json, benchmark.csv, wind-sensitivity.csv and saved mission files. All results are simulations generated for this report.

[L6] Browser evidence. Baseline, selected potato, cherry and WebODM CSV files and screenshots in figures/. Captured directly from the local application during report preparation.

Dataset attribution belongs to the original creators. The project name does not establish institutional authorship or endorsement. The report preserves the distinction between published dataset metadata, local integration choices and newly generated simulation outputs.

Appendix E Agricultural image atlas

E1 Apples and the scale of visual evidence



Figure 8. Six historical ADD256 patches, shown with their filenames and source split membership. Selection is illustrative rather than random; no detection overlay or prediction has been added [4, L2].

These patches make the agricultural task more concrete than the flight interface alone. The supplied images contain tree foliage, branches, background material and small visible fruit. Their native size is 256 × 256 pixels. Enlarging them on a report page improves inspection but does not add detail. A reader should therefore separate the physical scene, the original image resolution and the simulator's requested camera GSD.

The atlas illustrates why image availability and detection capability are different achievements. A useful fruit-counting study would need a defined counting target, a consistent rule for partly visible fruit and evaluation against the supplied annotations. This report does not calculate detection accuracy because no model predictions were produced. The simulator's canopy statistic also cannot be substituted for fruit count: it summarizes a vegetation proxy rather than individual objects.

Split membership is displayed to keep provenance visible. A future experiment should preserve the supplied training/test distinction and examine whether nearby patches originate from shared scenes before interpreting generalization. Every panel is mapped to its exact local source path in evidence/image-provenance.json. This mapping is more useful for verification than decorative stock photographs.

E2 Weed scenes and source masks

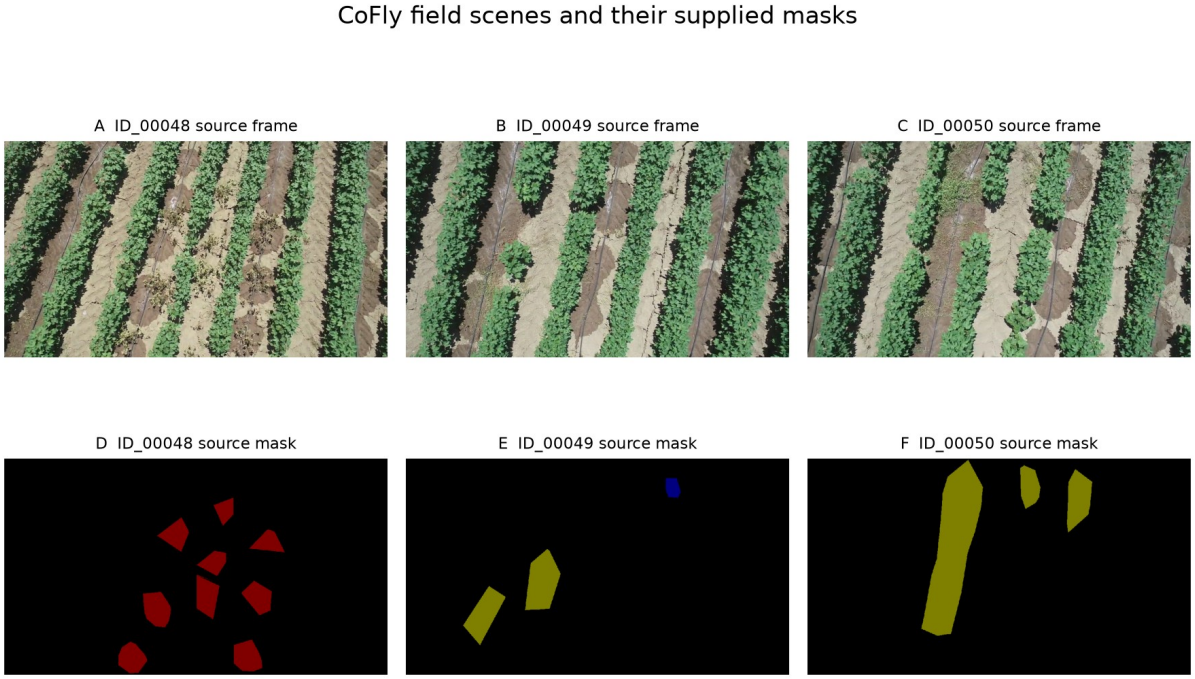


Figure 9. Three CoFly source frames in the top row and their paired supplied mask representations below. Columns preserve image–mask pairing. Display colours are those of the source representation; this plate does not infer a new class legend [5, 6, L2].

A paired frame and mask demonstrates the distinction between what a camera observes and what an annotation encodes. The frame includes crop, soil, illumination and other scene content; the mask encodes the dataset author's spatial labels. The project catalog connects these files rather than treating them as unrelated observations. This is why the 969 weed records cannot be described as 969 independent aerial scenes.

The proposed agricultural workflow would use the image to locate candidate weed regions and then compare those regions with the appropriate reference masks. Any future segmentation benchmark should state which label representation is used, how ignored pixels are handled and how performance is aggregated. A background-heavy image can make a naive pixel agreement percentage look favourable while still missing the small regions that matter to the survey objective. No such metric is claimed in the current simulation study.

These examples also explain why a fast survey alone is insufficient. An altitude that reduces mission duration may leave too little detail for the intended target. The present experiment imposes nominal GSD ceilings but does not test whether weed classes remain distinguishable at those ceilings. That question requires an image-quality experiment with known spatial scale and held-out labels.

E3 Cashew imagery and annotation scope

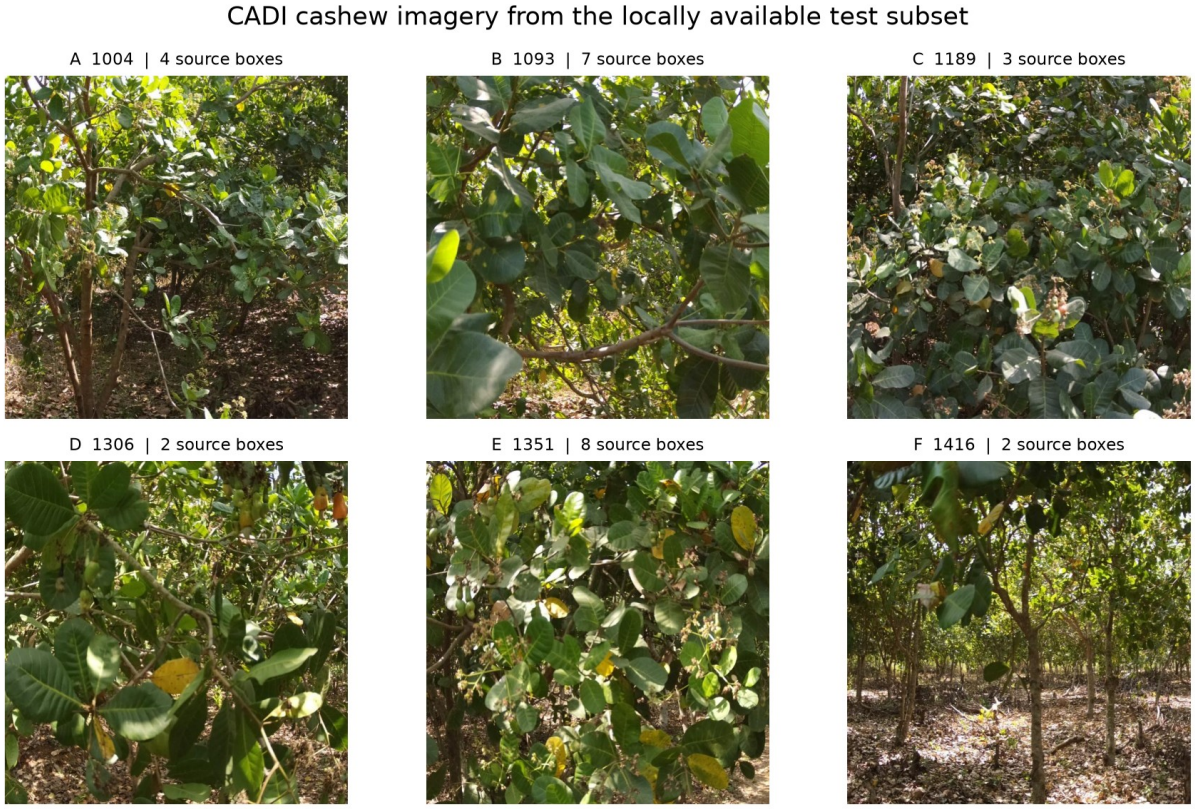


Figure 10. Six CADI source images from the 50-image local test subset. Panel titles show the source identifier and number of supplied box records; boxes are not rendered over the images and are not new predictions [L2, L4].

The cashew collection provides a different inspection problem from the orchard patches and aerial weed masks. Its supplied labels distinguish abiotic, insect and disease categories using normalized boxes. These categories come from the source documentation; this report does not visually diagnose plant conditions from the displayed examples. The small local subset is useful for testing image loading, label parsing and coordinate conversion.

A box count measures stored annotations, not severity, affected leaf area or independent biological cases. An image can contain several labelled regions, and similar source identifiers can occur in more than one file. A future analysis should examine related samples before creating a new split. Reporting a classifier score on these 50 images after tuning on them would not establish an independent test result.

This collection broadens the platform’s data-handling scope, but it does not increase the evidence supporting the potato flight optimum. The runtime comparison uses a different map and sampling function. The report therefore treats cashew imagery as evidence of supported agricultural content and annotation inspection, while reserving performance claims for the actual scenarios measured.

E4 Cherry ground photographs

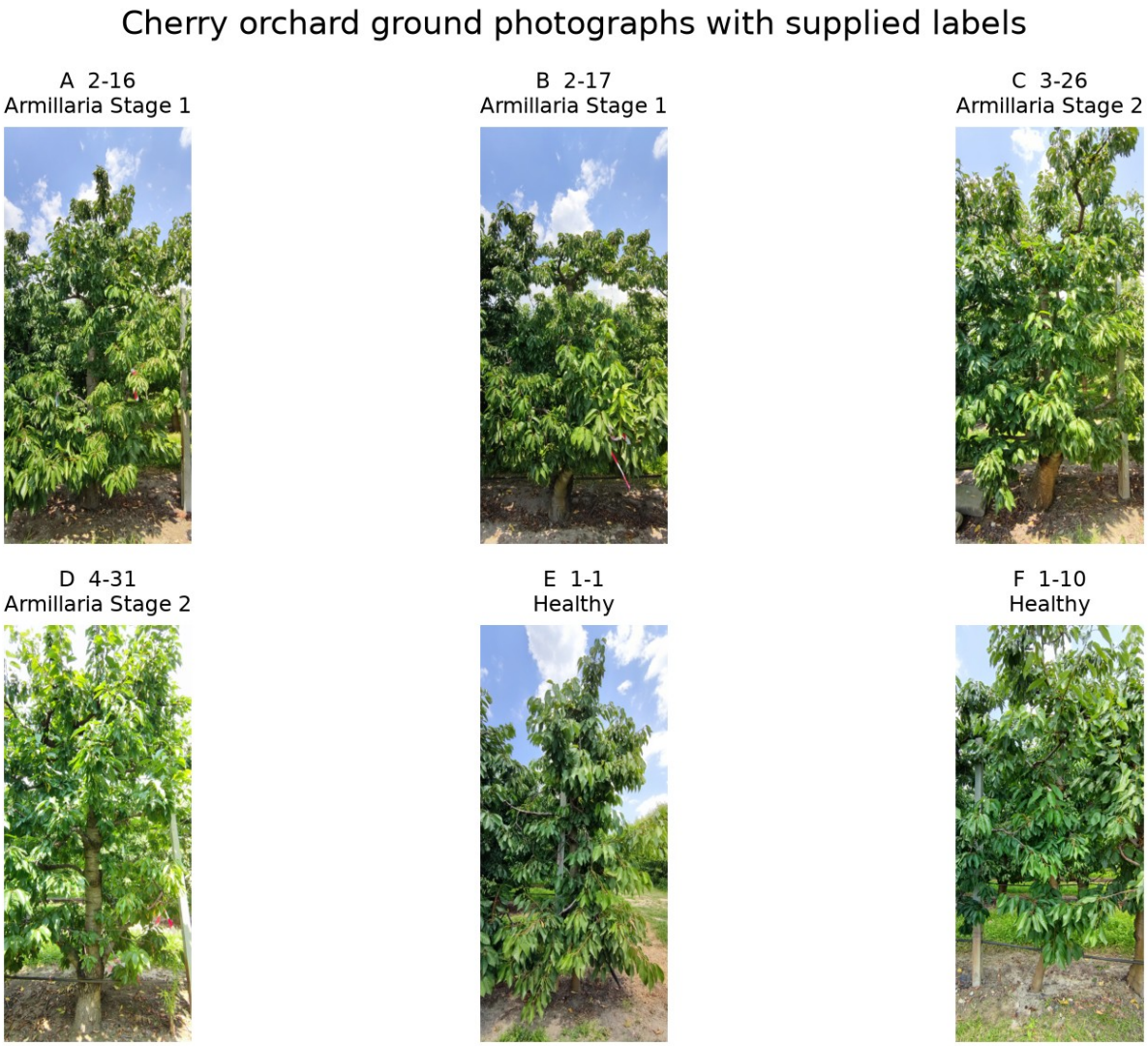


Figure 11. Six historical cherry ground photographs. Tree identifiers and class names are reproduced from the supplied catalog. These labels are source metadata, not diagnoses made by Drone Scout [L2, L4].

Ground photographs provide a view of individual trees that an overhead survey may not supply. They can retain visible context around branches, trunks and neighbouring vegetation. The catalog links these photographs with tree identifiers and corresponding sensor files, making it possible to inspect several representations of a source observation without implying that they were acquired by the current simulator.

The sample was chosen to expose available source categories rather than estimate their prevalence. Differences among panels may arise from view direction, lighting, camera distance, background and source condition. The photographs alone do not isolate the cause of those differences. A label associated with a tree also should not automatically be transferred to every aerial pixel over its surroundings.

To connect these records to a future drone study, the project would need verified correspondence between tree identity, location, acquisition date and the aerial footprint. The current workbook preserves source information but does not establish that spatial match. Until the association is checked, a simulated NDVI capture and a ground disease label should remain separate pieces of evidence. The library is valuable precisely because it keeps those identities available for later validation.

E5 Sensor bands and index displays

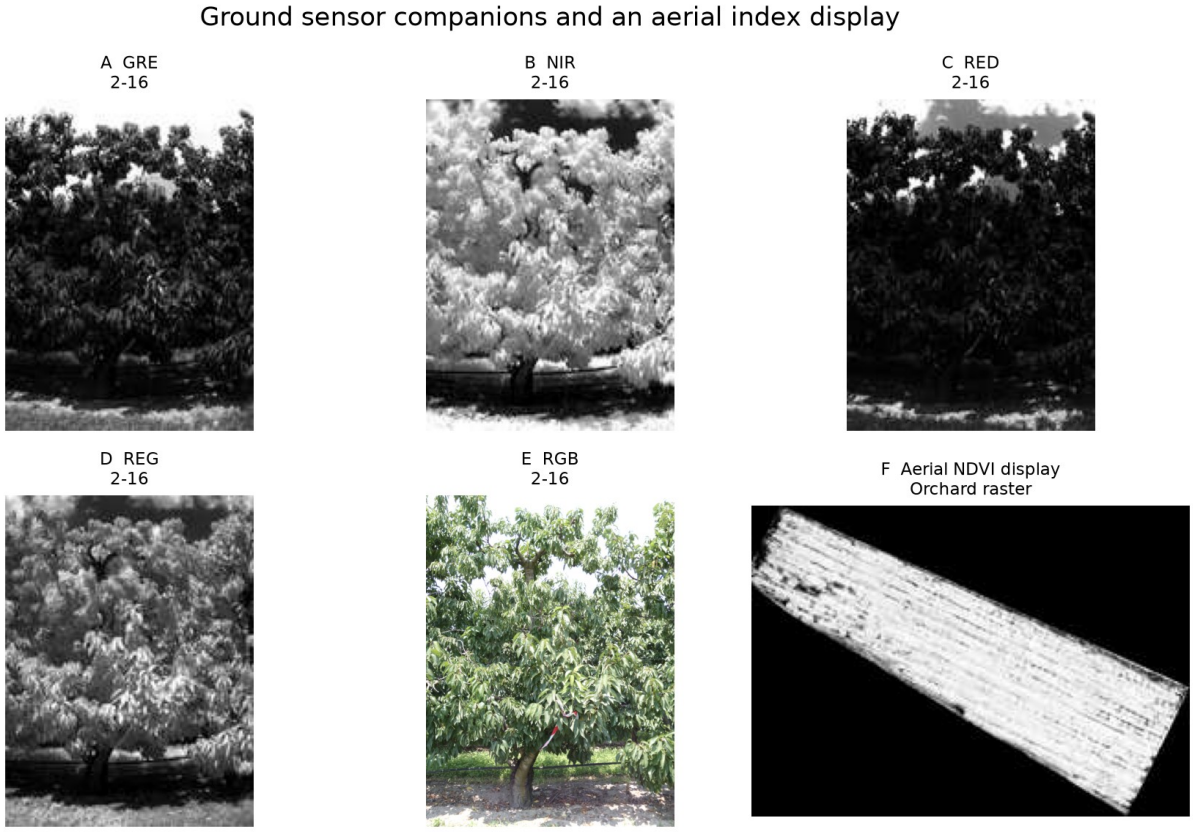


Figure 12. Five companion ground sensor records for tree 2-16 and one orchard NDVI display. The panels have different resolutions and display transformations. Brightness is not a shared calibrated scale; the aerial panel is not a pixel-aligned view of the ground photographs [L2, L3].

The green, near-infrared, red and red-edge records show why multispectral data require more care than ordinary colour thumbnails. Their stored values and display ranges are not interchangeable. The atlas uses the existing documented preview for floating-point rasters and preserves provenance for every panel. A visually bright region in one display is not, by itself, evidence of a quantitatively higher reflectance than a region in another display.

Equation 15 specifies the algebra of NDVI, but a meaningful application also depends on valid red and NIR measurements for corresponding locations. Band registration, no-data handling and radiometric preparation therefore belong in the interpretation, not merely the formula. The simulator samples a supplied derivative; it does not establish those upstream properties through a new calibration experiment.

The practical contribution is the ability to keep image, band, tree identifier and derivative relationships visible in one project. A future agronomic claim would require an explicit target and reference measurement, followed by an evaluation of how the index relates to that target. The present report supports inspection and planning; it does not establish an NDVI threshold for disease, irrigation or yield.

E6 Agricultural maps and spatial context

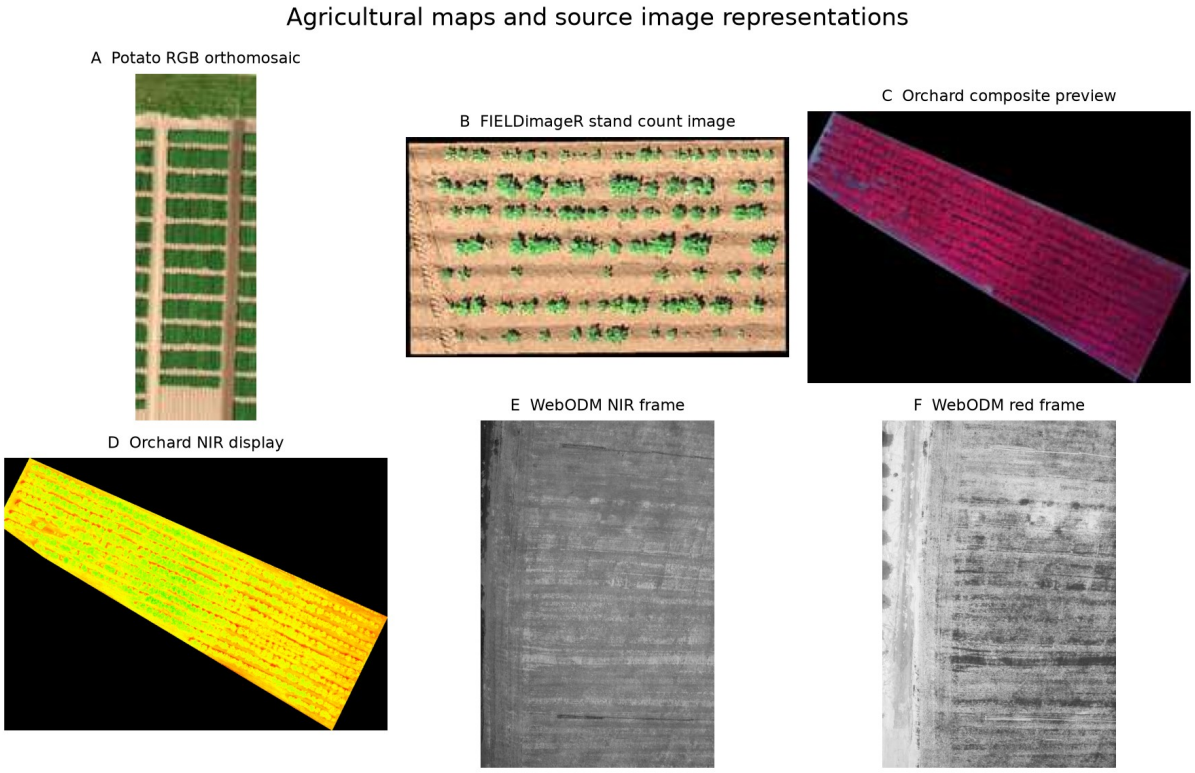


Figure 13. FIELDimageR potato and stand-count images, orchard composite and NIR displays, and paired WebODM NIR/red files. Some panels are compact catalog previews; the orchard composite uses the documented companion-source fallback. See the provenance manifest for exact files and display handling [L2, L3].

The potato orthomosaic makes the geometry of the benchmark visible. It is a narrow rectangular field, which favours lanes aligned with its long side. This shape is a central reason that the selected 40 m setting can cover the across-track extent in a single lane. Its advantage should not be generalized to a square field, irregular boundary or orchard with excluded areas without another experiment.

The plate also contrasts an orthomosaic with individual frames and display derivatives. These representations answer different questions. A mosaic gives a continuous scene representation; a spectral display emphasizes a particular stored band or index; paired camera frames retain individual acquisition identity. The application's simplified WebODM arrangement is not a substitute for a validated georeferenced reconstruction.

Images in this atlas are evidence of the project's available agricultural content, not evidence that every file contributed to every simulated mission. Potato, cherry and WebODM preset sampling use their own derivatives. The broader library supports inspection and configurable preview backgrounds. Maintaining this distinction prevents the total file count from being mistaken for the sample size of the flight study.

Appendix F Extended quantitative analysis

F1 Coverage is a selection constraint

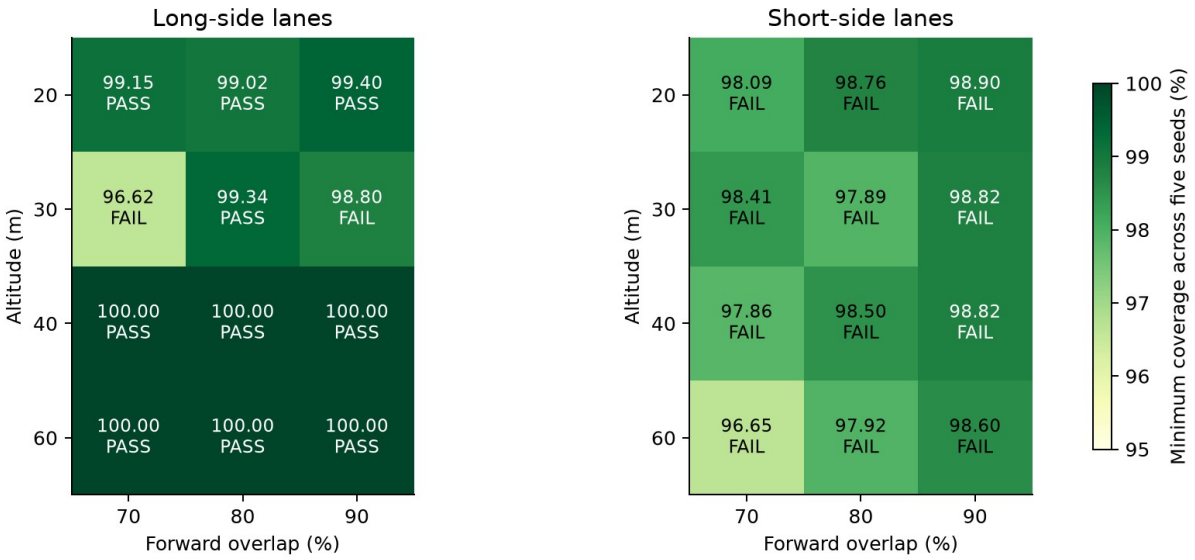


Figure 14. Worst-seed grid coverage for all 24 settings. PASS means at least 99% coverage in each of the five seeds; it does not include the separate GSD ceiling. Colour limits are fixed at 95–100%.

The heatmaps expose information hidden by a single average runtime. Each cell is the minimum coverage over five seeds, so a single poor outcome can disqualify a setting. This is deliberate: the study asks which fast settings also satisfy the stated coverage rule consistently within the tested seeds. A mean coverage statistic alone would conceal a failure if other runs compensate numerically.

At 30 m with 70% forward overlap and long-side lanes, the mean duration is shorter than the default, but minimum coverage falls to about 96.62%. The 40 m, 70% long-side group passes all five seeds and the 2.5 cm/pixel ceiling. The figure also shows that passing coverage is not enough to select a setting: 60 m scenarios have coarser nominal GSD and therefore fall outside the stated resolution limits.

The threshold is a study design choice. Changing it changes the feasible set and may change the preferred setting. Five seeds provide a small reproducibility check, not a reliable estimate of rare-event probability. Moreover, the underlying measure is a coarse touched-cell calculation. The next validation stage should examine exact footprint union, edge treatment and image usability before equating a PASS cell with a successful physical agricultural survey.

F2 Mission time energy and capture density

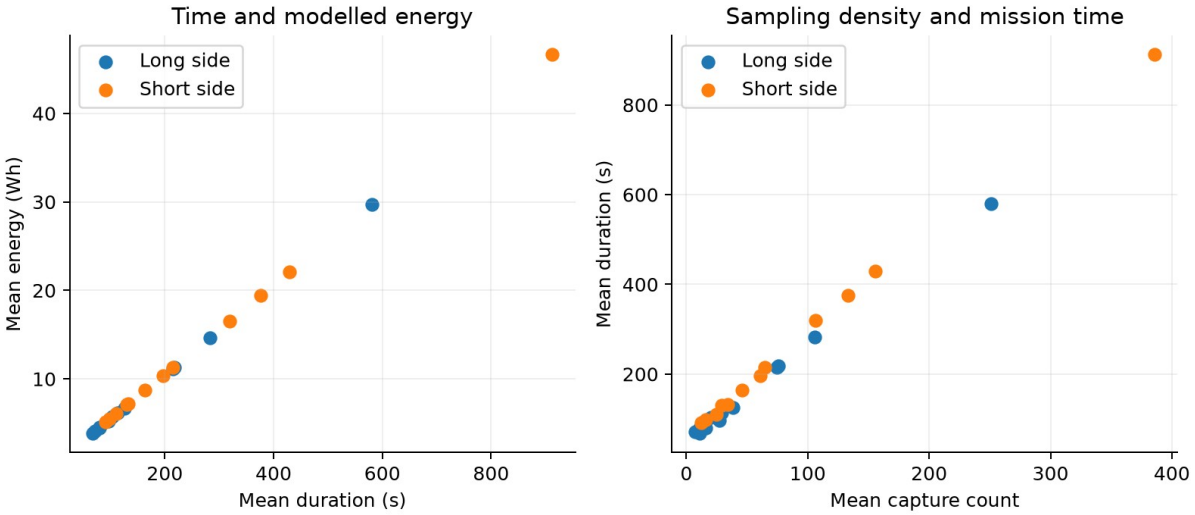


Figure 15. Setting means across five seeds. Left: simulated time versus energy from the implemented power model. Right: mean capture count versus time. Each point represents one setting, and colour distinguishes lane direction.

The association between time and energy is expected from the model. Equation 9 assigns a substantial continuous electrical load, and Equation 10 integrates that power over the mission. Longer missions therefore tend to consume more energy. This relationship is useful for checking internal consistency, but it is not independent experimental evidence for a real aircraft's power curve.

Capture count is also coupled to the planning inputs. Higher forward overlap reduces trigger spacing and constrains survey speed. More captures can therefore accompany a longer mission even though capture count itself is not directly entered as an energy penalty. Route length, turns and altitude further affect the relationship, which is why points do not define a universal seconds-per-image conversion.

The selected setting reduces mean captures from 38.2 to 10.8 relative to the default group. This supports an efficiency interpretation only for the stated coverage and GSD criteria. It does not show equivalent photogrammetric reconstruction, fruit visibility or robustness to blurred frames. If the intended downstream task needs additional overlap or redundant observations, the objective function should include those requirements before the fastest setting is selected.

F3 Variation across shared random seeds

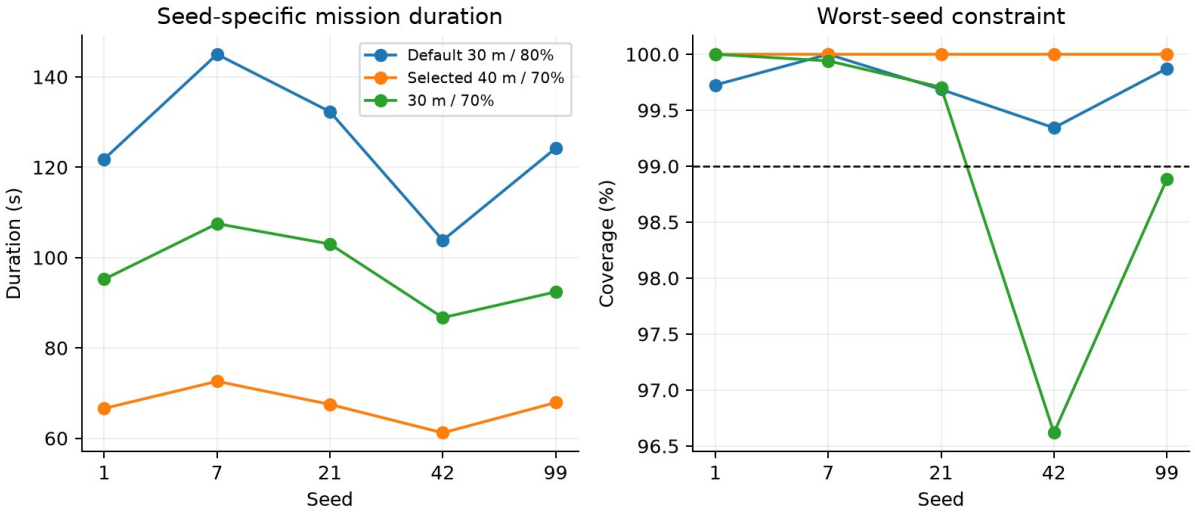


Figure 16. The same five seeds are compared across default, selected and faster-but-ineligible long-side settings. The dashed line marks the 99% coverage rule. Lines connect discrete seeds for readability; seed numbers have no physical ordering.

The default's mean duration of 125.42 s summarizes runs ranging from 103.8 to 145.0 s. Its seed-7 browser demonstration is therefore the slowest of those five default runs. Using that screenshot alone as a typical runtime would overstate the group mean. The selected group ranges from 61.2 to 72.6 s, and its seed-7 demonstration is likewise a particular realization rather than a universal outcome.

Shared seeds make configuration comparisons more controlled than choosing unrelated random settings for every run. However, the same seed does not guarantee identical disturbance histories at every spatial location: different route durations and state transitions can alter when random quantities influence the trajectory. The figure is best interpreted as a reproducible scenario comparison, not a matched physical field trial.

No confidence interval or significance claim is attached to these five-seed groups. The sample standard deviation describes variation in the recorded simulations only. A stronger robustness study would increase the seed count, predefine environmental ranges and repeat the evaluation on several field geometries. This would reveal whether the selected setting remains preferable when the narrow potato map is no longer the sole test case.

F4 Wind sensitivity and model interpretation

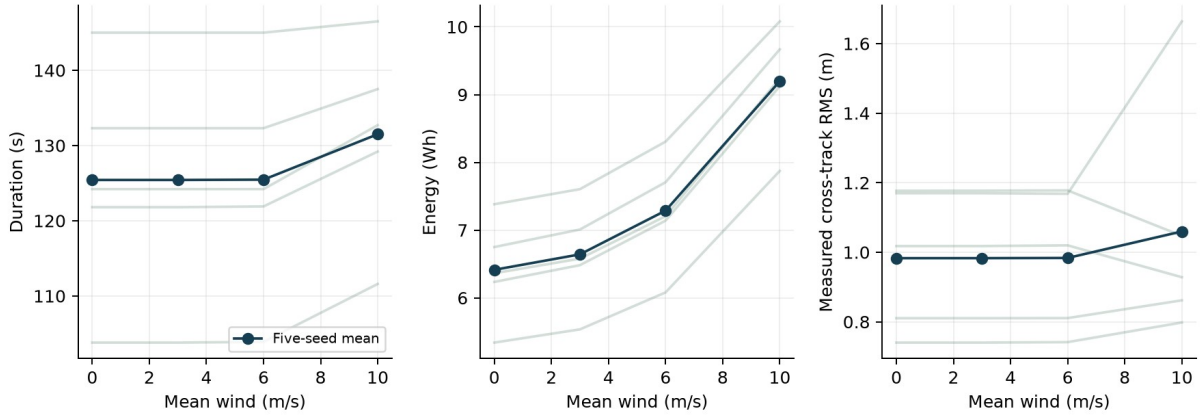


Figure 17. Twenty default-configuration wind runs. Faint lines show individual seeds; the dark line is the five-seed mean. Gust SD remains 0.8 m/s even when mean wind is zero.

Mean duration changes little from 0 to 6 m/s mean wind, whereas mean energy rises from approximately 6.414 to 7.288 Wh. At 10 m/s, mean duration increases to 131.50 s and mean energy reaches 9.197 Wh. These patterns follow from a controller that compensates for mean wind in desired air-relative velocity and a power function that increases with airspeed.

The near-flat time response at lower wind speeds should therefore be discussed as a consequence of the model's structure. It does not demonstrate that real agricultural flights are unaffected by wind. The simulator omits several mechanisms that could matter to image usability, including camera vibration, attitude-dependent footprints and motion blur. Its cross-track indicator also includes noisy measured position rather than independent truth tracking.

The faint curves show that averaging can hide seed-specific behavior. Even so, the experiment changes only one mean-wind magnitude parameter while keeping the default survey settings and other noise assumptions fixed. A future factorial study could vary wind direction, gust intensity and control limits together. Any such extension should state its scope before drawing conclusions about an operational wind envelope.

F5 Planner estimates and computing runtime

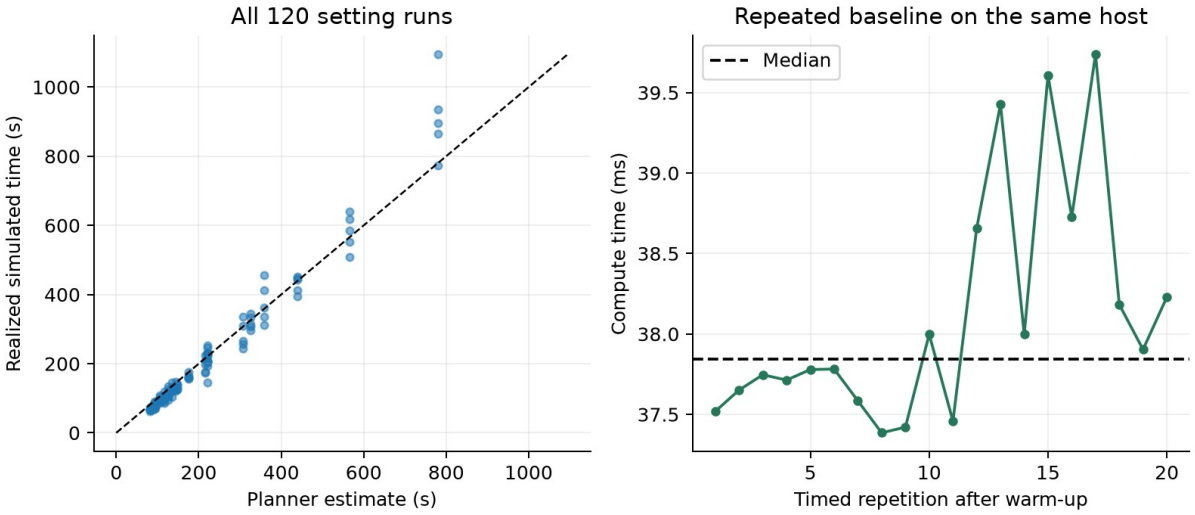


Figure 18. Left: all 120 realized mission times against planner estimates, with equality shown as a dashed line. Right: 20 warmed-up baseline compute repetitions on the recorded host, with their median. These two plots use different time units and measure different quantities.

The planner estimate is a compact approximation built from lane length, transit, turns, acceleration allowances and vertical travel. The realized simulation adds trajectory following, noisy position, capture gating and state transitions. Deviations from the equality line are therefore expected. The baseline's 134.97 s estimate and 145.0 s realization differ by 7.43%, but that one error should not be assigned to every configuration.

The right-hand plot answers a separate software-performance question. A median of 37.843 ms indicates how quickly this host executed the bounded headless baseline computation after warm-up. It does not mean that a physical drone can complete the 145 s mission in that time. Nor is it a browser frame-rate or interface-latency benchmark: rendering, loading and file writes were outside the timed scope.

The fastest measured compute repetition is 37.386 ms, while the maximum is 39.738 ms. Reporting both the median and observed range prevents the minimum from being presented as a guaranteed runtime. Host load, runtime version and implementation changes can alter these measurements. The accompanying frozen model and timing metadata make the experiment reproducible in method even when another computer produces different millisecond values.

F6 Capture footprints and spatial redundancy

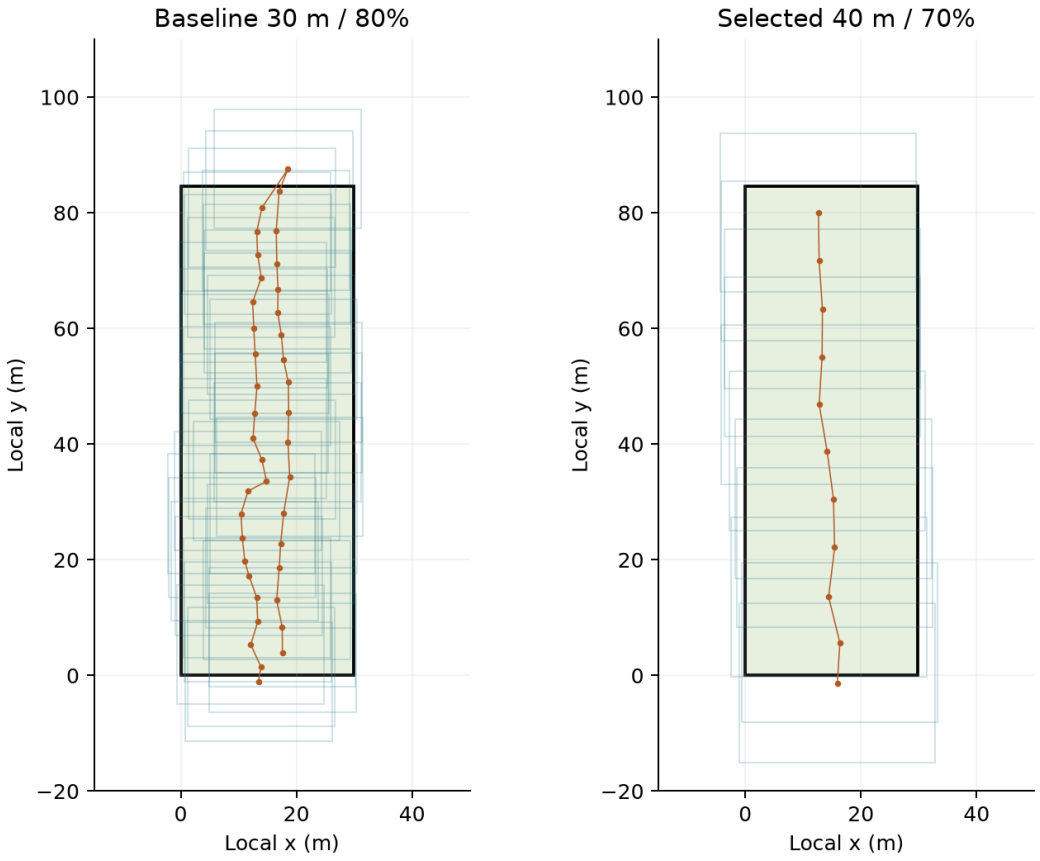


Figure 19. Seed-7 capture footprints in local map coordinates. The green rectangle is the potato map; blue outlines show exported capture bounds; orange points and connecting lines show capture locations and sequence, not the full aircraft telemetry path.

The baseline produces many overlapping footprints, whereas the selected setting uses fewer larger footprints. Both achieve 100% estimated touched-cell coverage in these seed-7 runs. The figure makes clear that this equality does not imply equal spatial redundancy. A single coverage percentage compresses the number, placement and degree of overlap of the observations.

Several footprints extend beyond the field. Their sampled values use only the source area available inside the map, which explains why data fraction can be below one even when cumulative grid coverage reaches 100%. In particular, a high canopy value at an edge capture characterizes the sampled portion; it is not a claim that the entire rectangular footprint contains vegetation.

The drawn rectangles are the simulator's axis-aligned local bounds. They do not encode camera perspective, terrain or actual aircraft attitude. A future spatial analysis could compute per-cell observation counts and an area-weighted field statistic, but such derived measures should be separately validated. This figure supports interpretation of the existing outputs without silently replacing the implemented coverage algorithm.

F7 Dataset composition and evidence weight

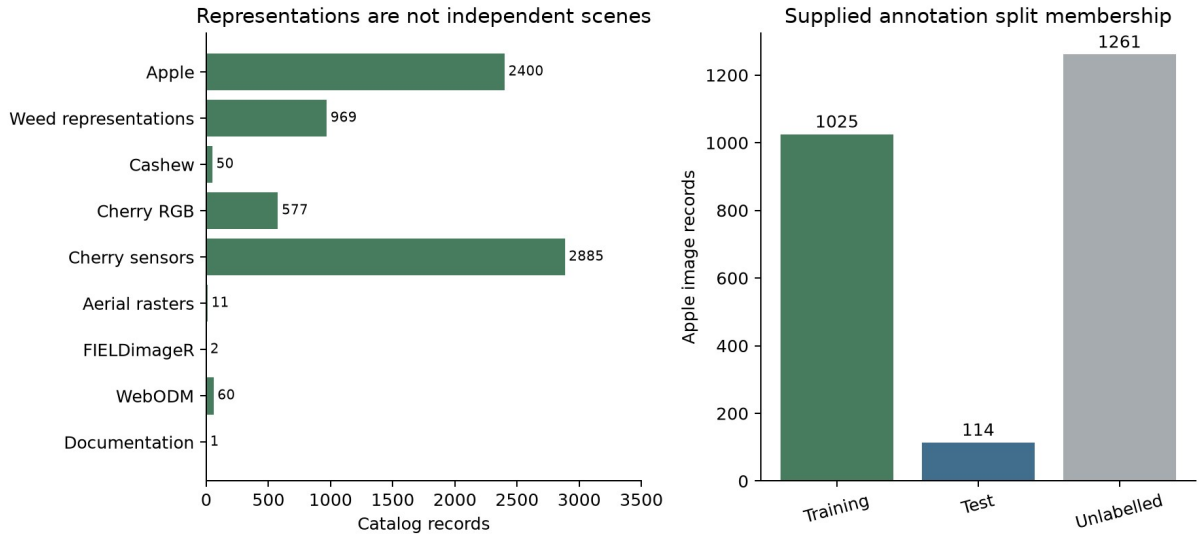


Figure 20. Left: all 6,955 catalog visual records, including one documentation image. Right: apple source split membership. Several collections include related bands, masks or derived views of the same observation [L2].

The sensor collection is the largest visual-record group because each of 577 ground observations has four TIFF band files and one RGB companion. Its 2,885 records therefore represent 577 related sensor groups, not 2,885 independent trees. Similarly, image, alternate mask and overlay files increase the weed catalog size without increasing the number of photographed scenes in the same proportion.

The apple split chart separates 1,025 training images, 114 test images and 1,261 unlabelled images. These numbers describe source membership, not successful model predictions. The catalog also preserves annotated examples with zero recorded fruit, which differ from unlabelled examples. Any learning pipeline should keep that distinction when constructing targets or reporting dataset size.

A data-rich interface is useful when it makes such relationships inspectable. It becomes misleading if the total record count is used as a proxy for biological diversity, training-set adequacy or field validation. The contribution demonstrated here is comprehensive access and provenance. Evaluating agricultural intelligence would require target-specific labels, meaningful independent units and a benchmark tied to the intended decision.

Appendix G Extended discussion

G1 What the project demonstrates

The project demonstrates an integrated path from heterogeneous agricultural source material to an inspectable survey simulation. Its strongest evidence is operational and reproducible: files are cataloged, relationships are retained, the planner generates missions, the simulator completes them, and browser exports agree with corresponding headless runs. The expanded atlas shows the variety of agricultural evidence available, while the new plots explain how mission outcomes change within the tested parameter grid.

The main engineering contribution is the connection between these capabilities. Dataset provenance informs what a sampled value means; flight geometry determines where that value is sampled; mission exports preserve enough configuration and telemetry to reproduce the outcome. A user can therefore inspect not just a map animation but the assumptions and source records behind the displayed metrics. This supports transparent experimentation without requiring a physical aircraft for every software iteration.

The performance result is conditional. The 40 m and 70% forward-overlap setting is best among the 24 tested configurations under the stated potato-map constraints. It is not an optimized general policy for all crops. Its one-lane advantage depends on field width relative to footprint width, and its lower image count may be unacceptable for a downstream task that needs finer detail or more overlap. The selected configuration is consequently an experimental finding with explicit boundaries, rather than a deployment recommendation.

The distinction between software consistency and external validity is central. Identical browser and headless CSVs show that two execution paths use the same underlying logic. They do not show that the power equation predicts a real battery, that the navigation noise matches a particular autopilot, or that canopy and NDVI values measure a desired crop outcome. Each of these would require a separate comparison against observations obtained independently of the simulator.

The image atlas also changes how completeness should be interpreted. Incorporating all available previews improves access and reveals material that was previously easy to overlook, but it does not mean that all source data have been quantitatively analyzed. A complete library and a complete scientific evaluation are separate milestones. This report now provides stronger visual evidence of the first and a bounded, traceable experiment for the second.

A defensible thesis claim is therefore that Drone Scout provides a provenance-aware agricultural survey simulation workflow and a reproducible demonstration of configuration trade-offs. Claims of a new disease detector, real-time physical sensing, validated yield prediction or globally optimal flight planning remain unsupported by the present evidence.

G2 From source imagery to an agricultural decision

An agricultural application ultimately needs a decision target. Counting visible fruit, locating weed regions, prioritizing tree inspection and monitoring vegetation cover require different observation scales and reference labels. The available collections support investigation of these tasks, but combining them in one library does not make their labels interchangeable. A future study should choose one target and define a measurable success criterion before selecting model architecture or tuning flight settings.

For fruit counting, the relevant unit might be a source patch, a tree or a complete orchard, and each choice changes the evaluation. For weed segmentation, the unit is a labelled spatial region with a specified mask convention. For tree condition, a supplied categorical label must be associated with the correct tree and acquisition time. In all cases, the mapping between a visual record and the intended biological unit should be explicit. Repeated bands and adjacent patches should not be silently treated as independent observations.

The current capture statistic is an intermediate measurement. The canopy sampler averages a historical derivative under a footprint. The cherry sampler averages a supplied index. Neither operation establishes a causal explanation for low values or a threshold for intervention. An agricultural decision layer would need a validated relationship between the sampled quantity and the target outcome, along with a way to represent missing data and uncertain observations.

Survey planning should then be evaluated against that outcome. The present objective minimizes mean time after applying coverage and nominal GSD constraints. A task-specific objective could additionally require a minimum usable image fraction, adequate target visibility, redundant coverage of priority regions or an acceptable error in an area-weighted estimate. These are proposed evaluation criteria, not capabilities that were measured in this report.

This leads to a practical research sequence: establish source identity and labels; define independent evaluation units; construct a transparent baseline; determine which spatial and image-quality conditions the target requires; and only then compare survey settings. The sequence keeps an attractive runtime improvement from being accepted when it undermines the information needed for the agricultural task. It also makes negative results interpretable: a fast setting may fail because of inadequate detail rather than poor route control.

The project already supplies useful foundations for that sequence through its catalog, source links, configurable survey geometry and export files. The missing work is empirical validation of the link from imagery to crop outcome. The expanded report makes this boundary explicit so that future collaborators can extend the platform without treating simulated coverage as a substitute for agricultural evidence.

G3 A validation programme for the next study

The next study should separate validation into source, geometry, dynamics and application stages. At the source stage, retain the existing manifest and verify image–label relationships on a documented sample. Check repeated records, source substitutions and split membership before creating training or test tables. Report both the number of visual files and the number of independent observation groups, because these are different denominators.

At the geometry stage, compare the touched-cell coverage implementation with an exact polygon-union calculation on simple synthetic fields and then on the potato footprint exports. Explicitly test partial edge cells, map gaps and out-of-bounds captures. This would quantify approximation error rather than relying only on visually plausible route plots. If per-cell redundancy is added, keep it separate from first-hit coverage and record its spatial distribution.

At the dynamics stage, use a known aircraft configuration and independent flight logs to compare takeoff time, speed profiles, route deviation, return behaviour and battery draw. Fit any empirical parameters on one set of flights and evaluate them on held-out flights. Record environmental conditions and acquisition settings alongside the logs. A close fit to a single flight would otherwise risk hiding compensating errors in power, speed and time assumptions.

At the application stage, evaluate one chosen agricultural target using its appropriate ground reference. Preserve source splits where supplied and group related observations so that the same scene or tree does not unintentionally inform both tuning and evaluation. Report errors in the units of the task, such as object count, spatial overlap or tree-level classification, only after actual predictions are generated. The current report supplies no baseline score for those future models.

The flight experiment itself should be broadened systematically. Add field shapes and scales, repeat the selected configuration under additional seeds, and vary wind direction and gust intensity as well as magnitude. Predefine which scenarios count as successful and how competing objectives are ranked. Publish per-run outcomes rather than only the fastest realization. If the optimum changes between fields, model that dependence instead of claiming a single universal altitude.

Success in the next phase would mean that a collaborator can trace an agricultural conclusion from its source observation through preprocessing, sampling, model output and evaluation. The current 120-run setting grid, 20 wind runs, frozen model and browser exports provide a reproducible starting point. The expanded figures add interpretive depth while preserving the central conclusion: the demonstrated result is a transparent simulation study with real historical agricultural content, and its physical and agronomic validity remain to be tested.